DOCUMENT REFERENCE

CLIENT	CAG - Jewel
PROJECT	Changi Airport Group – JEWEL
TITLE	**INTERFACE SPECIFICATION DOCUMENT – INTERFACES TXT FORMAT**
Section	**RETAIL**
CREATED DATE	03 NOVEMBER 2018
VERSION	1.7.6j

NEC Asia Pacific Pte. Ltd
80 Bendemeer Road
#05-01/02
Hyflux Innovation Centre
Singapore 339949
Tel: 6273 8333
FD365 FO: 6274 2226

DOCUMENT REVISION HISTORY

Version	Date	Changed By	Items Changed Since Previous Version
0.1	08-OCT-2018	Nicholas Fook	Initial creation
1.0	15-OCT-2018	Nicholas Fook	1st revision
1.1	25-OCT-2018	Nicholas Fook	2nd revision
1.2	03-NOV-2018	Nicholas Fook	Added Convention in Introduction. Revised Terminology in Introduction. Updated Tenant's Merchandise Classification diagram. Updated FILENAME & MODE in Page 16. Added CAG_CATG_CODE field in Tenant Product Hierarchy in Page 18. Updated the Remark for below fields in SKU Master at page 21 & 22. SKU_CATG_CAG PRO_FRDATE PRO_TODATE PRO_PRICE ITEM_ATTRIBUTE1 ITEM_ATTRIBUTE2 ITEM_ATTRIBUTE3 ITEM_ATTRIBUTE4 ITEM_ATTRIBUTE5 Update the Filename for PLU Master at Page 23. Added 3.4 Inventory Master Added 3.5 Promotion Master Amended the Character to Char. Amended & Added Record structure for CMD_CODE = 101. CMD_CODE = 102. CMD_CODE = 103.

Version	Date	Changed By	Items Changed Since Previous Version
			CMD_CODE = 106. CMD_CODE = 107. CMD_CODE = 108. Moved CMD_CODE = 109 up. Added CMD_CODE = 110. Amended & added fields for CMD_CODE = 111. CMD_CODE = 112. CMD_CODE = 131. CMD_CODE = 151. Added CMD_CODE = 152. Added Annex.
1.2a	05-NOV-2018	How Lee Kiau	Minor changes in format, Added COUPON_DESC, CAMPAIGN_DESC in CMD151 Added CR_STAT in cmd161 Refer to turquoise color
1.3	09-NOV-2018	Yulia	Add the following changes: • Added CAMPAIGN_GRP, CAMPAIGN_GRP_NAME in CMD151, 152. • Added "Price incl. tax" flag in SKU master. • Added "SO_RECEIPT_NMBR" in CMD1. • Added "SO_NMBR", "SO_INVOICE_NMBR", update "TRAN_STATUS" in CMD101. • Update "SALES_TYPE" in CMD111. • Few changes in CMD115. • Change payment name group from General to Cash, Added "PAY_ORIGIN" in CMD131. • Update on Annex – Payment Media Refer to **red font**.
1.4	22-NOV-2018	Yulia	Added the following changes: • Remove "Price incl. tax" flag in SKU master. • Add section 4.2 Summary of the EOD file content.
1.4a	26-Nov-2018	How Lee Kiau	**Revised Sign-Off version** Updated to version 1.4a from the previous version

Version	Date	Changed By	Items Changed Since Previous Version
			signed off on 5 Nov 2018 (Version 1.2a)
1.4b	1-April-2019	Yenty	Updated on the following changes CMD101 • Manual Receipt from M to O • Refund Receipt from M to O • Reason Code from M to O • Salesman Code from M to O CMD111 • Discount_Code from Char(15) to Char(20) • Item_Attrib1 to Item_Attrib15 from Char(20) to Char(60) CMD 151 • Discount_Info from Char(20) to Char(60)
1.4c	24-April-2019	Yenty	Updated on the following changes CMD111 • PLU from Char(20) to Char(80)
1.5	19-JUN-2019	Yong Jun LIM	Added new fields to: • SKU interface file • PLU interface file • Inventory interface file • EOD interface file CMD_CODE = 111 Edited 3.1 Tenant Product Hierarchy (Known as Category file) sample section by removing all the double quotes and adding comma in Landlord Category column. Added new interface file format: • Price interface file • Promotion Header interface file • Promotion Header Details interface file • Color Group interface file • Size Group interface file • Style Group interface file • Rental Department interface file CMD 106, CMD 107 • Added new columns: TEN, CAG, XML
1.6	1-JUL-2019	Yong Jun LIM	Master Files Format From Tenant's Backend System • Added new interface • Updated D365 TXT file requirement CATG Interface: • Added Note & Sample.

Version	Date	Changed By	Items Changed Since Previous Version
			SKU Interface: • Updated price related fields as NOT USE. • Added sample. PLU Interface: • Added sample. Price Interface: • Renamed from “Price Master” to “SKU Price Update” • Added inclusive tax and exclusive tax column. • Removed “From time” and “To time” column. • Added sample. Inventory Interface: • Renamed from “Inventory Master” to “SKU Inventory” • Added sample. Removed: • Promotion Header & Header Details Interface • Color Group, Size Group & Style Group Interface • Removed interfaces will be included in “CAG-ISD-Interfaces XML Formats” document Rental Department Interface: • Added Mode and Rental Description column • Added sample. CMD 151, CMD 152: • Added campaign related fields
1.7	9-JUL-2019	Yulia	Update the following: • Section 1 General overview • Section 2 General rules and guidelines • Section 4.2 Summary of EOD file content • Section 4.3 Update the sample content of the EOD file • Section CMD_CODE = 104 • Section CMD_CODE = 105 • Section CMD_CODE = 108 • Section CMD_CODE = 109 • Section CMD_CODE = 110 • Section CMD_CODE = 131 (additional fields) • Section CMD_CODE = 151 (additional fields)

Version	Date	Changed By	Items Changed Since Previous Version
			• Section CMD_CODE = 152 (additional fields) • Section CMD_CODE = 161 (additional fields) Update amount field from Numeric (8,2) to Numeric (20,2).
1.7.1	12-JUL-2019	Yulia	Update on CMD_CODE = 131 (additional fields related to voucher/payment option) Update on CMD_CODE = 111 (additional fields related to unique id)
1.7.2	19-JUL-2019	Yulia	Add Note in "SKU Price" (section 3.4) Update CMD 101 (add fields, remove DP, DF) Update CMD 111 (add fields for duty) Update CMD 107 (remove the Frequent flyer fields) Update CMD 109 (replace Date of birth with Age)
1.7.3	25-JUL-2019	Yulia	Add Price Unit in "SKU Price" (section 3.4) Update "unique id" no of character to 100 (CMD100).
1.7.4	06-AUG-2019	Yulia	Add "Discount" related information on CMD_CODE = 151 and 152
1.7.5	16-AUG-2019	Yong Jun LIM	Add Section 3 SFTP Folder structure Updated 5.3.15 Elapsetime formula Removed rental department interface and exchange rate interface.
1.7.5a	16-AUG-2019	Yong Jun LIM	Update CMD 1 (update SHOP_NMBR description). Add encryption related info under Section 5.3 Content of the EOD file. Update Section 3 SFTP Folder Structure.
1.7.6	16-AUG-2019	Yong Jun LIM	Update sample error log under Section 3 SFTP Folder Structure.
1.7.6a	21-AUG-2019	Derrick Lim	Updated point 12 of "2 General Rules and Guidelines". Updated grammar structure in document.
1.7.6b	13-SEPT-2019	Derrick Lim	Updated item attribute fields for SKU Master section.
1.7.6c	19-SEPT-2019	Derrick Lim	Updated section on Tenant Product Hierarchy, SKU Price Update
1.7.6d	19-SEPT-2019	Yong Jun Lim	Updated sample for section 4.4 Added null value definition for section 5.3.13 & 5.3.16.

Version	Date	Changed By	Items Changed Since Previous Version
1.7.6e	04-NOV-2019	Yong Jun LIM	Updated NOT USE fields from mandatory to optional.
1.7.6f	21-NOV-2019	Yong Jun LIM	Updated SKU_DISC description from "Determine if the product is discountable." To "Determine if cashier can apply manual discount on the product." Updated "N.A" to "NA" in SKU sample.
1.7.6g	17-JAN-2020	Yenty Elyanty	CMD 111 – Column SALESPERSON_CODE field type updated from CHAR(8) to CHAR(20) CMD 161 – Column CARD NO to show 19 char of Changi Rewards card number CMD 131 – Column PAYOPT_TYPE field type updated from CHAR(20) to CHAR(30) CMD 131 – Column PAYOPT_NAME field type updated from CHAR(60) to CHAR(150) CMD 104 – Column COLLECTION_STORE field type updated from CHAR(20) to CHAR(100)
1.7.6h	23-JAN-2020	Yenty Elyanty	CMD 111 – Column SALESPERSON_CODE field type updated from CHAR(8) to CHAR(20) CMD 161 – Column CARD NO to show 19 char of Changi Rewards card number CMD 131 – Column PAYOPT_TYPE field type updated from CHAR(20) to CHAR(30) CMD 131 – Column PAYOPT_NAME field type updated from CHAR(60) to CHAR(150) CMD 104 – Column COLLECTION_STORE field type updated from CHAR(20) to CHAR(100) CMD Add comment on encryption field length: Field maximum is 250 character, It will be encrypted up to 1000 char. This length (1000 char) will be applied for all fields in CMD= 102, 104, 106, 107, 109, 110
1.7.6i	23-JUL-2020	Yong Jun LIM	Updated CMD 111 TAX_CODE description and added new note referencing to annex 6.4. Added Annex 6.4 CMD 111 TAX_CODE matrix table.
1.7.6j	9-May 2023	Bhanu Prakash Alakanti	Updated Changi Pay payment line details

DISTRIBUTION LIST

Copy Number	Recipient
1	Project Technical Filing
2	CAG – JEWEL

ACCEPTANCE AND SIGNOFF

The contents herein comply with the contract documentation and the outcomes as discussed and agreed upon during the requirement gathering sessions.

This is to confirm the Interface Specification Document (ISD) has been accepted.

This version **1.7.6** of

Interface Specification Document – Interfaces TXT Formats

NEC		
Name	Designation	Signature & Date
Prepared By		
Nicholas Fook	Senior Consultant	
Yong Jun LIM	Functional Consultant	
Reviewed By		
Yulia	Solution Manager	
Jin Xiong Kwong	Project Manager	

CAG – JEWEL		
Name	Designation	Signature & Date
Reviewed & Approved By		
Derrick Lim	AM, Conc Ops, ACD	
How Lee Kiau	SM, CIT	

TABLE OF CONTENTS

DOCUMENT REVISION HISTORY **2**

DISTRIBUTION LIST **7**

ACCEPTANCE AND SIGNOFF **8**

TABLE OF CONTENTS **9**

INTRODUCTION **11**

ABBREVIATIONS 12
TERMINOLOGY 12
CONVENTION 14
LEGEND 14

1 GENERAL OVERVIEW **15**

2 GENERAL RULES AND GUIDELINES **16**

3 SFTP FOLDER STRUCTURE **17**

4 PRODUCT MASTER INTERFACE – GENERAL MERCHANDISE **19**

4.1 TENANT PRODUCT HIERARCHY (KNOWN AS CATEGORY FILE) 20
4.2 SKU MASTER 23
4.3 PLU MASTER 28
4.4 SKU PRICE UPDATE 29
4.5 SKU INVENTORY 31
4.6 PROMOTION MASTER 32

5 SALES TRANSACTION FILES **34**

5.1 NAMING CONVENTION 34
5.2 SUMMARY OF THE EOD FILE CONTENT 35
5.3 CONTENT OF THE EOD FILE 37
5.3.1 CMD_CODE = 1 (Required) *39*
5.3.2 CMD_CODE = 2 (Required) *40*
5.3.3 CMD_CODE = 101 (Required) *42*
5.3.4 CMD_CODE = 102 (Optional) *44*
5.3.5 CMD_CODE = 103 (Optional) *46*
5.3.6 CMD_CODE = 104 (Optional) *47*
5.3.7CMD_CODE = 105 (Optional) *48*
5.3.8 CMD_CODE = 106 (Optional) *49*
5.3.9 CMD_CODE = 107 (Optional) *50*
5.3.10 CMD_CODE = 108 (Optional) *51*
5.3.11 CMD_CODE = 109 (Optional) *52*
5.3.12 CMD_CODE = 110 (Optional) *53*
5.3.13 CMD_CODE = 111 (Required) *54*
5.3.14 CMD_CODE = 115 (Optional) *59*
5.3.15 CMD_CODE = 121 (Required) *61*
5.3.16 CMD_CODE = 131 (Required) *63*
5.3.17 CMD_CODE = 141 (Optional) *68*
5.3.18 CMD_CODE = 145 (Optional) *69*
5.3.19 CMD_CODE = 151 (Optional) *70*
5.3.20 CMD_CODE = 152 (Optional) *71*
5.3.21 CMD_CODE = 161 (Optional) *72*
5.3.22 CMD_CODE = 162 (Optional) *73*

6 ANNEX **74**

6.1 NATIONALITY 74
6.2 PAYMENT MEDIA 74
6.3 CAG/JEWEL PRODUCT HIERARCHY (SAMPLE GLOSSARY) 74
6.4 CMD 111 TAX_CODE MATRIX TABLE 74

INTRODUCTION

Through this document, NEC Asia Pacific Pte. Ltd. (hereby referred to as NEC) presents the business requirements for **CAG/JEWEL**.

The purpose of the document is to act as confirmations against the tender specification. It is the first phase of the Systems Development Life Cycle. During later phases, this document will be used to:

- Configure how the system will be setup
- Design how the solution will be delivered

The first phase of interface was created for Jewel which was implemented on March 2019. This document subsequently is enhanced for CAG phase to collect additional information.

Abbreviations

Abbreviation	Explanation
D365, D365FO	Dynamics 365 for Finance & Operations
POS	Point Of Sale
MPOS	Modern Point of Sale for D365
JCAD	Changi Airport Group - Jewel
CAG	Changi Airport Group
CVM	Centralized Voucher Management
CR	Changi Rewards
eV	eVouchers

Terminology

Please find below the list of terms where used in this document from the context of the implementation to convey the purposes to achieve the functionalities for Retail applications and not intended to be used as a legal definition.

Terms	Explanation
Landlord	CAG/JCAD lease out the respective premises within the Changi Airport Terminals/Jewel locations for the purposes of retail business operations.
Tenant	Parent Entity using the premises on lease from CAG/JCAD locations for retail store operations with 1 or more stores. Tenant entity will be able to share the employees, products, discounts, category hierarchy, etc within all the stores.
Stores	Each tenant will have at least 1 or more stores hosted within the premises of CAG/JCAD for the purposes of retail sales.
Retail Backend	Backoffice application access for the CAG, Tenants and administration users to access, view, manage, setup and maintain all information related to Retail POS operations. Microsoft D365 for Retail is the Retail backend.
Retail Admin	Retail backend administrator role user to maintain the Tenants, Stores, Registers, etc. backend configurations for the Jewel and CAG.
Interface	Transfer of information and data between applications in real time or scheduled interval using various predefined formats.
Product	SKU or Products or stocks refer to the items that are sold from the stores. Products may be item(stock) or service(non-stock)

Terms	Explanation
Category hierarchy	Tree view with multiple levels to display the parent-child relationship to the Product hierarchy maintained by CAG or Tenant.
Tenant Workspace	Tenant's self-service portal to access the Retail backend to maintain Tenant's master data and view transactions.
Tenant User	Tenants employees accessing the Tenant workspace to maintain and view information from Retail backend.
POS User	POS User or Employees or Workers or Staff, refer to the Tenant's store personnel created in Retail backend to access the POS.
Shifts	Shifts refer to the collection of POS transactional data and activities between two points in time. For each shift, the amount of money that is expected is compared against the amount that was counted and declared.
Close Shift	End of Shift and print Z Reading to receipt printer.
X Reading	Retail POS report printed anytime within the shift to display the collections and sales summary by each tender mode.
Z Reading	Retail POS report printed at the end of Close shift to display the final values with expected and declared values for the counted tender mode.

Convention

Field name	Explanation
Receipt Number	Below is the receipt number convention for reference: *Legal Entity Code + POS Id +YY+ddd+T+Running number*. XXXXXXX +YYJDY+T+99999 Legal Entity Code – 4 char [For e.g. WATS] POS id – 3 Char [For e.g. 121] Year – 2 Char [For e.g. 18] Jday – 3 Char [For e.g. 134] Transaction type – 1 Char [For e.g. 1] • 1 – Normal POS Sales • 2 – Sales order (Deposit) related transactions Running No. – 5 Char [For e.g. 00001] Maximum length = 18 chars Sample Receipt No. – WATS121181341**1**00001 This 18 digits numbers will be printed as receipt barcode.

Legend

Color	Explanation
GREEN	Modification made from existing CAG interface specification format.
YELLOW	Append or additional information from existing CAG interface specification format.

1 GENERAL OVERVIEW

This document defines the master interface tenant can send to the D365FO POS system deployed in CAG.

Tenant can maintain all the master data online on D365FO Backend system remotely through a web browser. Optionally, tenant may also import master files through SFTP and the data will be updated to POS system by predefined scheduler. The master data files which a tenant can send are as follows.

1) Tenant Product Hierarchy file (Category file)
- This file stores all category details in various level details.

2) SKU file
- This Stock Keeping Unit file stores all items that are available for sale at the shop.

3) PLU file
- This Price Lookup Unit file stores all PLUs with the associated SKUs. PLU can be used as a supplier bar code number or a second reference key. PLU information is optional at the POS.

4) SKU Price Update file
- This file stores price related fields.

5) SKU Inventory file
- This file stores the current stock on hand or adjustments to be made for items per store.

6) Promotion Master
- This file stores the tenant categories or products that are eligible in promotion.

7) Rental Department file
- This Rental Department file stores all the rental departments.

8) Currency Exchange Rate file
- This Currency Exchange Rate file is maintained by CAG and exchange rates are sent to all the POS system on regularly basis.

2 GENERAL RULES AND GUIDELINES

In order to import the master files correctly, CAG require all tenants to comply with the following General Rules and Guidelines.

1) All files to be sent to SFTP from tenant's system must be in ASCII format, data will be updated to POS system by predefined scheduler.

2) The value for each data field, regardless of the data type, are defined within a pair of double quotes.
   Note: For Jewel, D365FO accept data fields NOT wrapped in double quotes, which created issue when the data content is having comma or double quotes. (see #3 and #4)

3) Data fields that contain double quotation mark (") must be added with an additional double quotation mark (") and wrapped in double quotation mark.
   Example: Coffee Cup 2" → "Coffee Cup 2"" Blue Color"

4) Data fields that contain Comma (,) must be wrapped in Double quotation mark.
   Example: Boots, UGGS → "BOOTS, UGGS"

5) All fields are separated by Comma (,) as a delimiter.

6) Every record is terminated by the CR (carriage return) and LF (line feed) Chars.

7) Column headers (e.g. SKU_CODE, SKU_DESC) are not required in the interface file.

8) Validation checks will be performed during data import into D365FO system. If any records contain error(s), the record will be not be inserted/updated in D365FO while the rest of the "good" records will be processed. Erroneous records are logged in the system and recorded in the error log file (<original file name>.errorlog) stored in error folder located in SFTP server. Tenant should monitor and rectify the error record and resubmit to SFTP for reprocessing.

9) The width specified in the format is the maximum width allowed for each data field.
   Example of data format:
   Date - according to date format "YYYYMMDD"
   Time - according to time format "HH:MM:SS"
   Numeric - "25", "9842.23" (Note: No need to specify "25.00")
   Char - "iPhone XS Max"

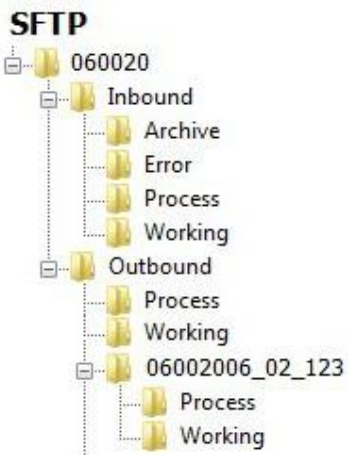
10) Char "M" at M/O column is used to indicate whether a field is a mandatory or optional field. All **mandatory fields' values MUST be provided**, else it will **result in import failure during import process**. If there is no data for a mandatory field, use following value according to data type: For Numeric - "0". For Char - "NA"

11) ~~Double quotation mark (") and comma(,) must not be used in anywhere in the data field.~~

12) Tenant's 1st batch of master file upload size has no file size limit. Within a specific timeframe after the 1st batch file upload, subsequent file upload size is limited to a maximum of 2MB (=/- 15,000 lines of record). Tenants are advised to split files with huge file size into smaller size. Tenants are advised to only provide incremental/delta records for subsequent file upload.

3 SFTP FOLDER STRUCTURE

This section defines the SFTP folder structure.



- Each tenant will be assigned with a SFTP folder that has a 6/7 digit Customer No. (e.g. '060020') as folder name.

- Each tenant SFTP folder will have 2 subfolders, "Inbound" and "Outbound" folder.

- "Inbound" folder contains 4 subfolders: "Working", "Process", "Archive" and "Error". The SFTP folder path need to be setup at Tenant Parameters.

SFTP folder	Definition	Access-able by tenant
<Tenant>/Inbound/Working	Tenant to access and provide master upload files (e.g. SKU, PLU & etc.). D365FO to retrieve these files to move them to Inbound Process folder.	Yes (read & write)
<Tenant>/Inbound/Process	D365FO to access & read the files for upload process. If upload successful, the file will be moved to Archive folder. If upload failure, the file will be moved to Error folder.	No
<Tenant>/Inbound/Error	Folder to keep the master files that has failed to process by D365FO. Tenant has the read permission to retrieve the file.	Yes (read only)
<Tenant>/Inbound/Archive	Folder to keep the master files that have successfully processed by D365FO.	Yes (read only)

- "Outbound" folder contains "Process" subfolder, "Working" subfolder and subfolders by stores with the format of <SAPCustCode><TerminalCode>_<UnitNo>
  Example: 06002006_02_123
  *Note: <SAPCustCode> is 6 digits, different from <Tenant> which is 6/7 digits

SFTP folder	Definition	Access-able by tenant
<Tenant>/Outbound/Working	Folder to keep the transaction files that processed by D365FO. Tenant to retrieve the files from this folder.	Yes (read & write)
<Tenant>/Outbound/Process	D365FO to access & write the transaction files (e.g. EOD). The file will be moved to Outbound Work folder when it is ready.	No
<Tenant>/Outbound/<Respective store folder>/Working	Folder to keep the transaction files by stores that processed by D365FO. Tenant to retrieve the files from this folder.	Yes (read & write)
<Tenant>/Outbound/<Respective store folder>/Process	D365FO to access & write the transaction files by stores (e.g. EOD). The file will be moved to Outbound Working folder when it is ready.	No

- Each tenant to own a login credentials to access the SFTP folders with either read or read & write permissions. **Please note that not all subfolders are access-able by tenant**.

- Sales and EJ files generated from D365FO will be encrypted. Sensitive fields (e.g. PDPA fields) in the sales / EJ files will be encrypted as well.

NOTE:
Only 1 SFTP folder & login credential will be assigned for each tenant.

- The files will be moved/transferred automatically by batch job upon processing of the interface files.

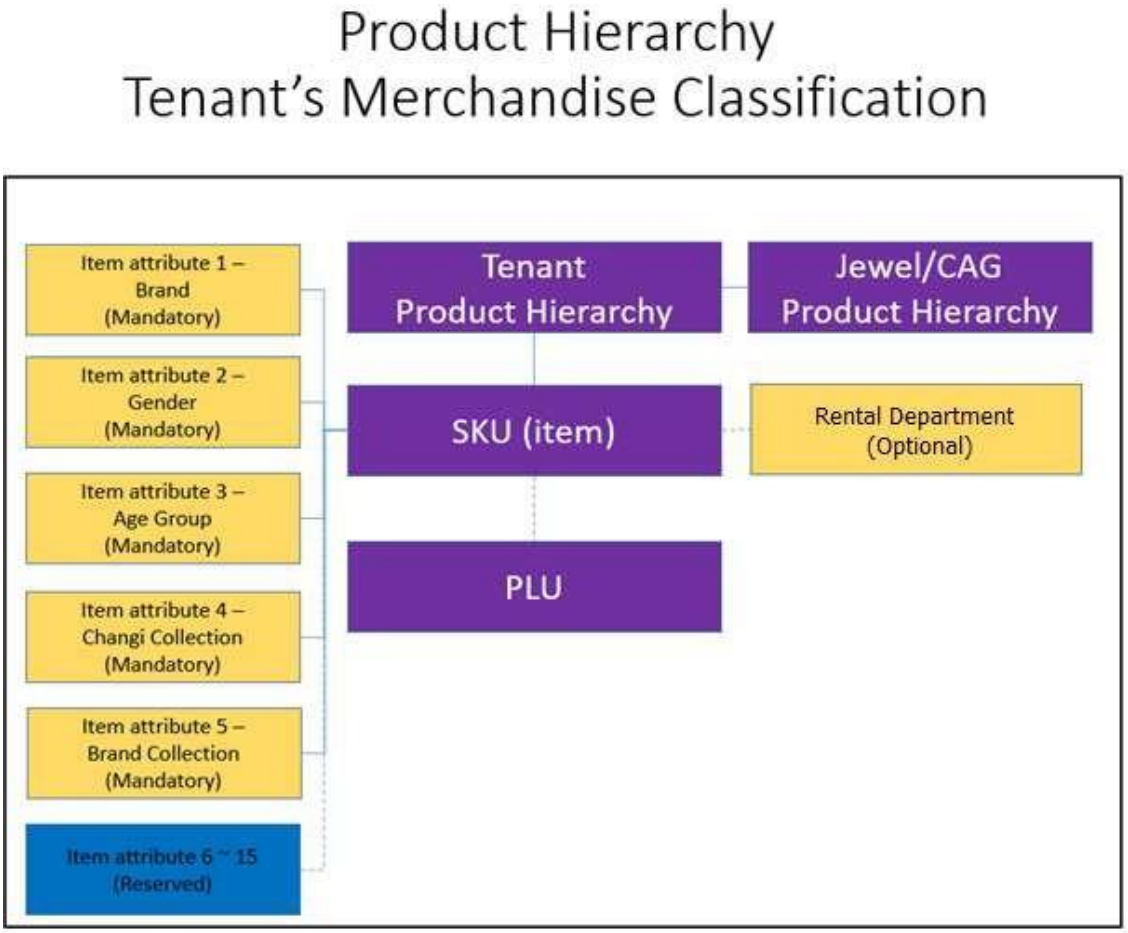
- D365FO will create 2 files for each master file that has failed to process in the Error folder.
  Example:
  SKU_<store ID>_<YYYYMMDDHHMMSS>.txt has failed to process.

  Files created:
  SKU_<store ID>_<YYYYMMDDHHMMSS>.Error which is a duplicate file of the original file.
  SKU_<store ID>_<YYYYMMDDHHMMSS>.ErrorLog which contain the error information for the respective data line submitted from original file.

- Example of the error log as below:
  Failed: Line 1 - CHILD_CATG_CODE not found
  Failed: Line 3 - Mandatory fields are not filled: SKU_CODE
  Failed: Line 4 - PLU code duplicated, must be unique
  Accepted: Line 10 - SKU_DESC is truncated, exceeded maximum 60 Characters
  Accepted: Line 10 - Invalid Rental Department

4 PRODUCT MASTER INTERFACE – GENERAL MERCHANDISE

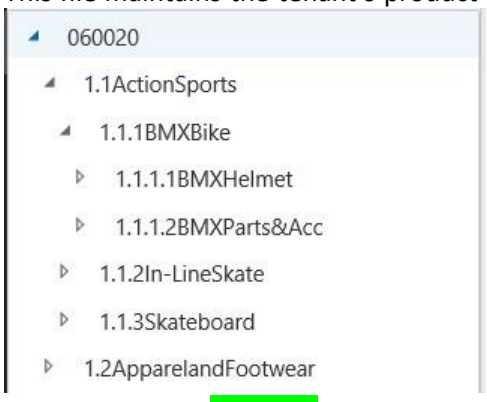
The section defines the GM Master Interfaces.



4.1 Tenant Product Hierarchy (Known as Category File)

Tenant product hierarchy defines the tenant's product categories across all stores operating in Jewel/CAG premises. This first level of the hierarchy is the Tenant ID, Tenant ID is the 6 digits customer code assigned by Jewel/CAG. Each category code MUST be unique.

This file maintains the tenant's product category in a hierarchy format as below:



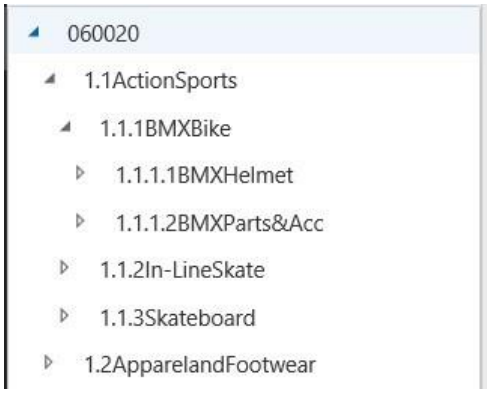
Filename: **CATG_<tenant>_<YYYYMMDDHHMMSS>.txt**

Note: <tenant> is the 6/7 digits customer code assigned by Jewel/CAG.

Sno	Field Name	Type	M/O	Remark
1	PARENT_CATG_CODE	Char (20)	M	Parent category code in the tenant product hierarchy. NOTE: This first level of the hierarchy is the Tenant ID. Tenant ID is the 6/7 digits customer code assigned by Jewel/CAG.
2	CHILD_CATG_CODE	Char (20)	M	Child category code in the tenant product hierarchy. This category code must be unique. NOTE: Numbering is not mandatory.
3	CHILD_CATG_DESC	Char (60)	M	Name of child category in the tenant product hierarchy.
4	CAG_CATG_CODE	Char (20)	M	Jewel/CAG category code in the Jewel/CAG product hierarchy. Product hierarchy list will be provided by Jewel/CAG. This mapping facilitates joint tenant and landlord promotions for selected categories, tenant products that are mapped to CAG category can be configured in the system for participation in the promotion, without intervention from tenants.

				For example: Purchase $50 of Skin Care (Based on CAG category) products in Changi Airport, get 1 hr free parking. NOTE: This field is required only for the lowest level of CHILD_CATG_CODE. For more details refer to Annex – CAG hierarchy

Sample:



060020,1.1ActionSports, Action Sports
060020,1.2ApparelandFootwear, Apparel and Footwear
1.1ActionSports,1.1.1BMX,BMX Biking,
1.1ActionSports,1.1.2In-LineSkate,In Line Skating,
1.1ActionSports,1.1.3Skateboard,Skateboard,
1.1.1BMX,1.1.1.1BMXHelmet,BMX Helmets,**OTHER ACCESSORIES**
1.1.1BMX,1.1.1.2BMXParts&Acc,"BMX Parts, Accessories", **OTHER ACCESSORIES**

Note #1: The category hierarchy start with the <tenant>. <tenant> is the 6/7 digits customer code assigned by Jewel/CAG. For above example **060020** is the <tenant>.

Note #2: Numbering (example 1.1, 1.1.1, 1.2 above) is for illustration only and not mandatory in the category code (provided that it is unique), you may remove the numbering to reduce the number of characters, as the category code have an input limit of 20 characters.

Note #3: Items highlighted in blue are the category codes and items highlighted in purple are the description of the category codes. Both the category codes and category description are mandatory fields during importing of the category master file.

Note #4: CAG_CATG_CODE is only required to be mapped against lowest level of CHILD_CATG_CODE. For above example only **1.1.1.1BMXHelment** and **1.1.1.2BMXParts&Acc** are the lowest level (no further breakdown), hence they are required to be mapped to a CAG defined category code. (**OTHER ACCESSORIES**).

Note #5: For data field containing comma (,), please wrap it with double quotation mark (“). This is to tell system the data field is one field and not separated by comma. For above example “BMX Parts, Accessories” is one field.

4.2 SKU Master

This file stores all products & its' pricing available for sales in Jewel/CAG premises.

For tenants who have multiple stores that carry different product assortments (e.g: DFS carry watch, fashion & liquor & tobacco products sold in different type of concessions), tenants have the option to send the SKU master to specific or all stores from the Retail backend.

However, for tenants who have multiple stores that carry the same product assortment, but with different rental department code, tenants must send multiple SKU master file for the same product assortment. The SKU master file will need to be assigned to the store that has a rental department code.

Eg. For the same group of product assortments, 1 SKU master file must be created for products without rental department code, 1 SKU master file must be created for products with rental department code.

Note: All the SKU price related fields have a dedicated SKU Price Update interface.

Filename: **SKU_<store ID>_<YYYYMMDDHHMMSS>.txt**
Note: <store ID> is the 5 digits code provided by Retail POS.

Sno	Field Name	Type	M/O	Remark
1	FILENAME	Char (8)	~~M~~ O	NOT USE
2	MODE	Char (1)	M	Valid values are ~~F/~~A/E/D This field is the mode of operation '~~F': Full refresh~~. 'A': Add SKU • If SKU_CODE already exist, system will skip this record • Price related fields will be ignored 'E': Edit SKU related information only. • Price related fields will be ignored 'D': Delete SKU • The record will be removed from POS, but will not be deleted in Retail backend if there is transaction history. If record does not exist, no action will be taken.
3	SKU_CODE	Char (16)	M	Unique item number
4	SKU_DESC	Char (60)	M	Description of the product. Only up to first 40 Chars will be printed on the POS receipt.
5	COSTPRICE	Numeric (20,2)	O	Cost of the product in local currency (SGD).
6	AVGCOST		O	NOT USE
7	SKU_PRICE	Numeric (20,2)	~~M~~	NOT USE

Sno	Field Name	Type	M/O	Remark
			O	Price update will be using SKU Price Update Interface ~~Base price of the product.~~ ~~For taxable stores (JCAD, landside), the price should be inclusive of tax.~~ ~~For non-taxable stores (airside), the price should be exclusive of tax.~~
8	SKU_PRICE1	Numeric (20,2)	O	NOT USE ~~New selling price of the product.~~ ~~For taxable stores (JCAD, landside), the price should be inclusive of tax.~~ ~~For non-taxable stores (airside), the price should be exclusive of tax.~~
9	SKU_EDATE1	Date	O	NOT USE ~~Effective date of the new selling price~~
10	SKU_ETIME1			NOT USE
11	SKU_CATG_TENANT	Char (20)	M	Lowest level of category code of tenant product hierarchy. It could be tenant product subclass level.
12	SKU_CATG_CAG	Char (20)	O	NOT USE. ~~To follow the CAG_CATG_CODE defined in Tenant Product Hierarchy.~~
13	SKU_ROL	Numeric (9,2)	O	NOT USE
14	SKU_ROQ	Numeric (9,2)	O	NOT USE
15	SKU_QOH	Numeric (9,2)	O	NOT USE
16	TAX_CODE	Char (1)	M	'G': Taxable (E.g: 7%) 'N': Non-taxable
17	PRO_FRDATE	Date YYYYMMDD	O	NOT USE. ~~Date from which the promotional price is used.~~
18	PRO_FRTIME			NOT USE. ~~Promotion to be configured in D365FO retail back office by tenant.~~
19	PRO_TODATE	Date YYYYMMDD	O	NOT USE. ~~Last date which the promotional price is used.~~
20	PRO_TOTIME			NOT USE. ~~Promotion to be configured in D365FO retail back office by tenant.~~
21	PRO_PRICE	Numeric (20,2)	O	NOT USE. ~~Promotion price to be used for the promotional period~~

Sno	Field Name	Type	M/O	Remark
22	OPEN_ITEM	Char (1)	M	Y/N. Determine if the product is an open item. If 'Y', selling price is defaulted and POS to prompt cashier to key in a new price.
23	KIT		O	NOT USE
24	REMARKS			NOT USE
25	SKU_DISC	Char (1)	M	Y/N. ~~Determine if the product is discountable.~~ Determine if cashier can apply manual discount on the product.
26	SKU_PRICE_LALO			NOT USE
27	SKU_PRICE_HALO			NOT USE
28	MIN_AGE			NOT USE
29	ITEM_ATTRIB1	Char (20)	M	This is an item attribute for brand (To be defined by tenants).
30	ITEM_ATTRIB2	Char (20)	O	This is an item attribute for gender. Please strictly follow values stated below. Values: 'MALE', 'FEMALE', 'UNISEX' *This field is mandatory for selective tenants in the fashion concession.
31	ITEM_ATTRIB3	Char (20)	M	This is an item attribute for age group. Please strictly follow values stated below. Values: 'ADULT','CHILD<12', 'ALL'
32	ITEM_ATTRIB4	Char (20)	M	This is an item attribute for Changi collection. Please strictly follow values stated below. Values: 'CHANGIEXCL', 'LOVESG', 'TRAVELEXCL', 'TRAVELEXCL_LOVESG', 'CHANGIEXCL_LOVE SG', 'TRAVEL_CHANGI EXCL,'NA'.
33	RENTAL_DEPT	Char (20)	O	This is an item attribute to be defined by tenant that have been mutually agreed by Jewel/CAG. The purpose of the field is for rental structure computation based on product categorization.
34	USE_STOCK	Char (1)	M	Y/N If 'Y', determine the product is stocked item. NOTE: Once transaction exists for the item, then this value can no longer be changed.

Sno	Field Name	Type	M/O	Remark
35	SKU_LONG_DESC	Char (1000)	O	This is the long description of the product for searching purpose.
36	BlOCK_SALES	Char (1)	M	Y/N If 'Y', determine the product to be blocked from selling.
37	ITEM_ATTRIB5	Char (20)	O	This is an item attribute for brand collection (To be defined by tenants) (E.g: series, limited edition). E.g: Lewis Hamilton X T H, Adidas NMD, GoPro Hero, iPhone
38	ITEM_ATTRIB6	Char (20)	O	This is an item attribute for Language. Please strictly follow values stated below. Values: 'ENGLISH', 'CHINESE', 'ALL OTHERS','NA' *This field is mandatory for selective tenants in the book concession.
39	ITEM_ATTRIB7	Char (20)	O	This is an item attribute for Genre. Please strictly follow values stated below. Values: 'K-COS', 'J-COS', 'ALL OTHERS','NA' *This field is mandatory for selective tenants in the perfume & cosmetics concession.
40	ITEM_ATTRIB8	Char (20)	O	This is an item attribute for Geography. Please strictly follow values stated below. Values: 'LOCAL', 'OVERSEAS' *This field is mandatory for selective tenants in the travel communication and iShopChangi concession.
41	ITEM_ATTRIB9	Char (20)	O	Reserved for future use.
42	ITEM_ATTRIB10	Char (20)	O	Reserved for future use.
43	ITEM_ATTRIB11	Char (20)	O	Reserved for future use.
44	ITEM_ATTRIB12	Char (20)	O	Reserved for future use.
45	ITEM_ATTRIB13	Char (20)	O	Reserved for future use.
46	ITEM_ATTRIB14	Char (20)	O	Reserved for future use.
47	ITEM_ATTRIB15	Char (20)	O	Reserved for future use.
48	ZERO_PRICE_VALID	Char (1)	O	Y/N. If 'Y', means system allows zero sales price. If leave blank, means 'N'
49	SEARCH_NAME	Char (20)	O	Item search/short name.
50	ENABLE_UNIQUEID	Char (1)	O	Y/N. If 'Y', determine the product to Capture Unique Identifier (mandatory) upon POS

Sno	Field Name	Type	M/O	Remark
				sales transaction with the info code (pop up screen), e.g. serial number (trigger info code to capture). If leave blank, means 'N'

Sample:

,A,10001,BMXHelmet,,,100,,,,1.1.1.1BMXHelmet,,0,0,0,G,,,,,0,N,N,,Y,0,0,0,BMX,NA,NA,NA,,N,,N,BMX,,,,,,,,,,,,

,A,10002,BMXPartsandAccessories,,,20,,,,1.1.1.2BMXParts&Acc, ,0,0,0,G,,,,,0,N,N,,Y,0,0,0,BMX,NA,NA,NA,,N,,N,BMX,,,,,,,,,,,,

4.3 PLU Master

This file contains all PLU codes with the associated SKU codes. When a product code is entered at the POS system (manually or through bar code scanner), the system will first search the PLU Master file for the code. If it is not there, it then searches the SKU Master file.

Note:
Multiple PLU codes can be made to associate to the same SKU code.

Filename: **PLU_<tenant>_<YYYYMMDDHHMMSS>.txt**

Sno	Field Name	Type	M/O	Remark
1	FILENAME	Char (8)	~~M~~ O	NOT USE
2	MODE	Char (1)	M	Valid values are A/D/E This field is the mode of operation. Values: 'A' or 'E': Add / Update record. If record already exists, the record will be updated. If record does not exist, the record will be added. 'D': The record will be deleted. If record does not exist, no action will be taken.
3	PLU_CODE	Char (80)	M	PLU or EAN barcode of the product.
4	SKU_CODE	Char (16)	M	SKU code of the product. The SKU code must exist in SKU Master file.

<u>Sample:</u>
,A,100012468,10001
,A,100021357,10002

4.4 SKU Price Update

This file stores the product's pricing. The product pricing file is not per store specific, it will apply for all stores respectively whether it's inclusive or exclusive of tax.

Filename: **PRICE_<tenant>_<YYYYMMDDHHMMSS>.txt**

Sno	Field Name	Type	M/O	Remark
1	MODE	Char (1)	M	Valid values are A/D This field is the mode of operation 'A': Add / Update record. If record already exists, the record will be updated. If record does not exist, the record will be added. 'D': The record will be deleted. If record does not exist, no action will be taken.
2	SKU_CODE	Char (16)	M	Unique item number restricted to 16 characters due to D365 limitation.
3	STORE_ID	Char (5)	O	Store ID. Leave this blank if price is not by store.
4	PRICE_INCLTAX	Numeric (20,2)	M	Sales price inclusive of tax.
5	PRICE_EXCLTAX	Numeric (20,2)	M	Sales price exclusive of tax.
6	PRICE_UNIT	Numeric (8)	O	Price unit (e.g. price $1 for price unit 100 is equal to price = $0.01) NOTE: Blank/Zero (0) will be imported as one (1).
7	PRICE_FRDATE	Date YYYYMMDD	M	Effective start date of the selling price.
8	PRICE_TODATE	Date YYYYMMDD	M	Effective end date of the selling price. For original price, default as 20991231, so that it never expired.

NOTE:

When the **same SKU has more than 1 price with overlapping periods**, the **cheapest price will take effect**.

Sample #1: Current active price & promo price

Item 10001 with current active price of $12.20 (exclusive tax) & promo price of $10.00 (exclusive tax) starting from 8th July to 15th July 2019.

A,10001,,13.05,12.20,,20190701,20991231
A,10001,,10.70,10.00,,20190708,20180715

Sample #2: SKU price by store

Item 10002 with current active price of $7.99 (inclusive tax) for store 80123 and $6.99 (inclusive tax) for store 80124.

A,10002,80123,7.99,7.47,,20190701,20991231
A,10002,80124,6.99,6.53,,20190701,20991231

For tenants who intend to submit 2 prices, price inclusive and exclusive tax that have 2 different prices. Tenants can set the tolerance level betwee both prices. Price inclusive tax will only be applied to outlet's POS that are configured for Landside.

There are 2 components (ratio tenant and calculated ratio tenant) calculated before reaching to tolerance.

Price inclusive tax = $ 159
Price exclusive tax = $ 129

1. Calculated ratio tenant = (Sales tax against to price exclusive tax + Price exclusive tax) / Price exclusive tax = (9.03 + 129) / 129 = 1.07
2. Ratio tenant = (Price inclusive tax / Price exclusive tax) = 159/129 = 1.2325

Tolerance = abs (Calculated ratio tenant - Ratio tenant) = abs(1.07 - 1.2325) = abs(-0.1625) = 0.16.

Please note – abs is used to remove the sign from a negative numeric value. And it always show positive number.

4.5 SKU Inventory

This file stores the current stock on hand or adjustments to be made for items per store.

Filename: **INVDETAILS_<store ID>_<YYYYMMDDHHMMSS>.txt**

Sno	Field Name	Type	M/O	Remark
1	FILENAME	Char (8)	~~M~~ O	NOT USE
2	SKU_CODE	Char (16)	M	Unique item number
3	ACTION	Char (16)	M	Values: 'Add': Add inventory quantity 'Subtract': Subtract inventory quantity 'Update': Overwrite quantity on hand captured in the system
4	INV_VALUE	Numeric (9)	M	Inventory quantity in +ve value.

Sample:

,10001,Update,10
,10002,Update,5

4.6 Promotion Master

This file stores the tenant category or SKU eligible for promotions. Promotion Header interface is only available in XML format due to the complexity of the discount configuration, please refer to "CAG-IDS-Interfaces XML Formats" document for more information.

Pre-requisite: Tenants are required to create the type of promotion & mechanic as a Discount Header in D365FO before the file can be uploaded.

Filename: **PROMO_<tenant>_<YYYYMMDDHHMMSS>.txt**

Sno	Field Name	Type	M/O	Remark
1	DISC_ID	Char (20)	M	Unique promotion/discount assigned from D365FO. This field should follow the same Discount ID that has assigned in Discount Header.
2	TENANT_CATG_CODE	Char (20)	O	Tenant category that is eligible in promotion. It is mandatory if SKU_CODE is undefined.
3	SKU_CODE	Char (16)	O	Product that is eligible in promotion. It is mandatory if TENANT_CATG_CODE is undefined.
4	LINE_TYPE	Char (10)	M	Indicate if tenant category or product to be include/exclude from promotion. Values: 'Include','Exclude'
5	DISC_METHOD	Char (20)	M	Method of the discount. Values: 'AmountOff': Discount amount off. 'PercentOff': Discount % off. 'Price': Discount price. Note: Only 'PercentOff' & 'Price' are supported for 'MixAndMatch' discount with *Line specific* configuration.
6	DISC_VALUE	Numeric (20,2)	M	Discount value that correspond to the DISC_METHOD.
7	M&M_LINEGRP	Char (1)	O	This field correspond with MixAndMatch line group that is defined in Discount Header. It is mandatory for 'MixAndMatch' discount.

Sample:

Simple Discount

SD0001,,10001,Include,PercentOff,10, ← Item 10001 is included in the promo
SD0001,TOPS,,Include,AmountOff,5, ← Category "TOPS" is included in the promo
SD0001,,10002,Exclude,AmountOff,0, ← Item 10002 is excluded from the promo

Mix and Match Discount

MM0001,TOPS,,Include,,,A ← Category "TOPS" is included in the promo (Group A)
MM0001,BOTTOMS,,Include,,,B ← Category "BOTTOMS" is included in the promo (Group B)
MM0001,,20001,Exclude,,,B ← Item 20001 is excluded from the promo (Group B)

Quantity Discount

QD0001,,10001,Include,,, ← Item 10001 is included in the promo
QD0001,,10002,Exclude,,, ← Item 10002 is excluded from the promo

Threshold Discount

TD0001,,10001,Include,,, ← Item 10001 is included in the promo
TD0001,,10002,Exclude,,, ← Item 10002 is excluded from the promo

5 SALES TRANSACTION FILES

This file contains the detailed data of every transaction issued by each POS. This means that if a shop opens 3 POS on a particular day, 3 files should be generated for that day. 1 file will be generated for each POS on a business day. Sales file will be generated provided that they login to the MPOS.
This file will be auto-generated immediately after end of day (EOD) is performed.

5.1 Naming Convention

The file naming convention is as follows:

SALES_ aaaaaaaaaaa _pppppp_xxxx_YYMMDDHHMM.txt

SALES	Prefix and it denotes the file is a sales transaction file
aaaaaaaaaaa	Tenant ID from where the file is generated. E.g: 0600200329008.
pppppppp	POS ID from where the file is generated. E.g: 0000101,
xxxx	Transaction file number starting from 0001. Every time a file is generated for the same POS, it should be incremented by 1. This number should be reset to 0001 on reaching 9999.
YYMMDDHHMM	Date and time the file is generated
txt	File format in “txt”

Example of file name:

(i) SALES_0600200329008_0000101_0014_0407192130.txt

NOTE:

The file name cannot contain any of the following Chars: -

\ / : * “ ‘ |

5.2 Summary of the EOD file content

Below is the summary of the EOD file content:

Command Code	Description	Once/ Multiple records	Mandatory/ Optional
1|OPENED	Beginning of the file	Once per file	Mandatory
2|	'No Sale' related functions	Can be multiple within a file	Optional
101|	Header of each transactions	Once per receipt	Mandatory for each receipt
102|	Tenant additional customer info	Once per receipt	Optional
103|	Customer master info	Once per receipt	Optional
104|	Order information for pickup/collection scenario	Once per receipt	Optional
105| 105|	Sales order delivery mode at item level	Multiple per receipt (one record represent one receipt line)	Optional
106|	Information about customer's passport	Once per receipt	Optional
107|	Information about customer's boarding pass	Once per receipt	Optional
108|	Customer email address for e-receipt	Once per receipt	Optional
109|	Non-traveler information	Once per receipt	Optional
110|	Customer related info identified by CAG	Once per receipt	Optional
111| 111|	Information of item sold including void items	Multiple per receipt (one record represent one receipt line)	Mandatory for each receipt
115| 115|	Footer record of a deposit transaction	Can be multiple per receipt (e.g. for customer order cancellation, if there's deposit refund ***and*** deposit forfeit in one receipt)	Mandatory for customer order, except for customer return.
121|	Footer record of a transaction that stores Tax, Service Charge and Discount	Once per receipt	Mandatory for each receipt

Command Code	Description	Once/ Multiple records	Mandatory/ Optional
131|T 131|C	Currency code and amount tendered or changed or rounding adjustment	Can be multiple per receipt 131|T is for payment (amount received) 131|C is for change/refund (amount issued)	Mandatory for each receipt (If zero sales (free item), system will send 131|T as CASH with zero (0) amount.
141|	Footer record that stores tax details	Can be multiple per receipt, if two tax codes are used for the items sold	Optional
145|	Footer record that stores amount from unpaid accumulated transaction	Can be multiple per receipt	Optional
151|	Footer record of discount transactions & used coupon/campaign info	Can be multiple per receipt	Optional
152|	Footer record of coupon/campaign giveaway (receipt)	Can be multiple per receipt	Optional
161|	Changi rewards information	Once per receipt	Optional
1|CLOSED	Closing of the file	Once per file	Mandatory

5.3 Content of the EOD file

The file naming convention is as follows:

The file is a multi-record-type and variable-record-length file. The first field (REC_TYPE) indicates what type of record it is, the general format is as follows:

REC_TYPE|Data_field_1| Data_field_2|…….|Date_Field_n<CR><LF>

1) All fields should be in ASCII format. The field delimiter is | (the pipe Char).

2) The width specified is the maximum possible width for the field. However, actual number of Chars stored may be less than the maximum. Null values (nothing between 2 pipes for any middle field, or nothing between the last pipe and the record terminator for the last field) means that user did not enter any value (or the POS did not generate it) to the fields while issuing the receipt.

3) Each record is terminated by the CR (carriage return, Hex 0D) and LF (line feed, Hex 0A) Chars for DOS or Windows systems and LF for UNIX systems.

4) The "Optional" field stated in the record type (ie. cmd 2) means the record can be omitted if the POS system does not capture such information. The record should be generated in the interface file if POS captures such info.

5) Field Value with carriage return or pipe (|) character will be removed from the EOD file.

6) Sales and EJ files generated from D365FO will be encrypted. Sensitive fields (e.g. PDPA fields) in the sales / EJ files will be encrypted as well.

7) **In the customer pax field (MPOS), system allows tenant to mask or encrypt sensitive data, e.g. passport. This information is stored in CMD=102, 104, 106, 107, 109, 110. If tenant decides to encrypt a particular field, the field length might exceed the original field length defined in this document. (e.g. encrypting 12 characters passport might become 30 to 50 characters). Tenants are advised to decrypt those fields before storing in the database. Alternatively, tenant may consider extending the field length to store the encrypted data.**

   Field maximum is 250 character, It will be encrypted up to 1000 char. This length (1000 char) will be applied for all fields in **CMD= 102, 104, 106, 107, 109, 110**

The following is a sample transaction file for illustration purpose. The actual detailed definition of each CMD record type is given in subsequent sections.

CMD_CODE

Sample Data Fields

```
1|OPENED|2400200078123|121|100043|0010|20181123|23:20:15|1278|20181123|200010
2|SIGNON|10|20181123|07:32:38|1225|N|0.00|100051|
2|NOSALE|10|20181123|07:46:22|1225|N|0.00|100052|
101|100044|10|20181123|07:50:26|1225||||1227|||N|SALE||100053||
111|81301|1.00|320.90|320.90||G|M2_5003|64.18|C and P|HANDBAGS|||239.93|20.00|%|16.80|8801051704938|0.00|0.00|0.00|
BAGS|Yellow Snakeskin Bag|BAGS|HANDBAGS|WHOO|FEMALE|ADULT|N.A|1227|ART_SALE||||||||||||||
111|81302|1.00|320.90|320.90||G|M2_5003|64.18|C and P|HANDBAGS|||239.93|20.00|%|16.79|8801051704939|0.00|0.00|0.00|
BAGS|Yellow Snakeskin Bag|BAGS|HANDBAGS|WHOO|FEMALE|ADULT|N.A|1227|ART_SALE||||||||||||||
121|513.44|33.58|0.00|0.00|33.59|E|N|M6_9999|0.00|6.54|0.00|0.00|2.00|513.45|Y|73|
131|T|CASH|SGD|1.0000000|1000.00|||1000.00||||||N|||||CASH||||
131|C|CASH|SGD|1.0000000|486.55|||486.55||||||N|||||CASH||||
141|G|33.59
101|100045|10|20181123|08:45:22|1225||||1222|||N|SALE||100056||
111|81301|1.00|320.90|320.90||G|M2_5003|64.18|C and P|HANDBAGS|||239.93|20.00|%|16.80|8801051704938|0.00|0.00|0.00|
BAGS|Yellow Snakeskin Bag|BAGS|HANDBAGS|WHOO|FEMALE|ADULT|N.A|1222|ART_SALE||||||||||||||
121|256.72|16.79|0.00|0.00|16.80|E|N|M6_9999|0.00|6.54|0.02|0.00|1.00|256.73|Y|85
131|T|CASH|SGD|1.0000000|300.00|||300.00||||||N|||||CASH||||
131|C|CASH|SGD|1.0000000|43.25|||43.25||||||N|||||CASH||||
131|C|ADJUST|SGD|1.0000000|0.02|||0.02||||||N|||||ADJUST||||
141|G|16.80
161|0000200199000065463901001|510|0|Y
2|SIGNOF|10|20181123|08:46:35|1225|N|0.00|100058|
2|SIGNON|10|20181123|08:46:47|1222|N|0.00|100059|
101|100046|10|20181123|10:25:19|1222||||1222|||N|SALE||100060||
111|81303|1.00|83.20|83.20||G|M1_0099|8.32|C and P|HANDBAGS|||69.98|10.00|%|4.90|8801051783667|0.00|0.00|0.00|BAGS|
Blue Snakeskin Bag|BAGS|HANDBAGS|WHOO|FEMALE|ADULT|N.A|1222|ART_SALE||||||||||||||
121|74.88|4.90|0.00|0.00|4.90|E|N|GSTPROMO|0.00|6.54|0.02|0.00|1.00|74.88|Y|62
131|T|CASH|SGD|1.0000000|100.00|||100.00||||||N|||||CASH||||
131|C|CASH|SGD|1.0000000|25.10|||25.10||||||N|||||CASH||||
131|C|ADJUST|SGD|1.0000000|0.02|||0.02||||||N|||||ADJUST||||
141|G|4.90
161|0000200100001346775030 02|420|0|Y
1|CLOSED|2400200078123|121|100046|0010|20181123|23:20:15|1278|20181123|200010
```

Data for transaction 100044

Data for transaction 100045

Data for transaction 100046

NOTE: All fields which type = Char, special characters is accepted, e.g. ,~!@#$%^&*()_+=-`[]\{};'":/.,<>?

5.3.1 CMD_CODE = 1 (Required)

This record contains the information of the shop, POS terminal number and timestamp when this file was opened / closed. One record with 'OPENED' status appears at the beginning of the file to indicate the beginning of the transaction file for a business date. One record with 'CLOSED' status appears at the end of the transaction file to indicate the ending of the transaction file for a business date.

Record Structure

Sno	Field Name	Type	M/O	Remark
1	FILE_STAT	Char (6)	M	File status: - OPENED - CLOSED
2	SHOP_NMBR	Char (20)	M	Tenant ID is assigned by CAG/Jewel. Currently 13 digits. E.g. 2400200078123 The Tenant ID will change as per tenancy agreement changes.
3	POS_NMBR	Char (6)	M	3 digits CAG POS ID
4	RECEIPT_NMBR	Numeric (10)	M	Opening / closing receipt number depending on FILE_STAT; Which applies for POS Sales Transactions. "Opening receipt number" is the last receipt number issued by this POS just before the last EOD "Closing receipt number" is the last receipt number issued by this POS just before the current end-of-day. It will become the "Opening receipt number" of the next EOD; It uses the CAG Transaction running number. For D365FO retail POS to send the Transaction type (1 char) + Running No. (5 chars) as RECEIPT_NMBR
5	TRAN_FILE_NMBR	Numeric (4)	M	Transaction file number, same as the one in filename
6	DATE	Date YYYYMMDD	M	Date when this file is generated. Note that this date is for information only and it is **NOT** the business date.
7	TIME	Time HH:MM:SS	M	Time when this file is generated.
8	USER_ID	Char (20)	M	If FILE_STAT = OPENED, this is the initial of the staff ID who did previous POS EOD processing if FILE_STAT = CLOSED, this is the initial of the staff ID who did the current POS EOD processing.
9	SALE_DATE	Date YYYYMMDD	M	Business date where all the transactions belong to

Sno	Field Name	Type	M/O	Remark
10	SO_RECEIPT_NMBR	Numeric (10)	M	Opening / closing receipt number depending on FILE_STAT; Which applies for Sales Order related transactions. "Opening receipt number" is the last receipt number issued by this POS just before the last EOD "Closing receipt number" is the last receipt number issued by this POS just before the current end-of-day. It will become the "Opening receipt number" of the next EOD; For D365FO retail POS to send the Transaction type (1 char) + Running No. (5 chars) as RECEIPT_NMBR

5.3.2 CMD_CODE = 2 (Required)

These record stores the date and time for each time a 'No Sale' function is done. These details are recorded in this file for analysis only. For example, how often cash drawer is opened with this function.

Record Structure

Sno	Field Name	Type	M/O	Remark
1	OPERATION	Char (6)	M	is the NON-SALE operations being done; Example (not exhaustive): NOSALE, CANCEL, SIGN ON , SIGN-OFF, EOS & EOD NOSALE – Open cash drawer without sale CANCEL – Transaction void SIGNON – POS logon SIGNOF – POS logoff EOS – Close shift XREAD – Print X ZREAD – Reprint Z RPRINT – Reprint Z EOD – Retail statement is posted
2	SHIFT_NMBR	Numeric (8)	M	Shift number from D365FO POS. If login without open shift, default to 0.
3	DATE	Date YYYYMMDD	M	Date when the operation is performed.
4	TIME	Time HH:MM:SS	M	Time when the operation is performed.
5	USER_ID	Char (20)	M	Staff ID who performed the operation.
6	TRAINING	Char (1) Y/N	M	Flag to determine whether training mode is on / off. For non production environment, this will be 'Y'.
7	AMOUNT	Numeric (20,2)	M	Amount of voided transaction. Mandatory if OPERATION = CANCEL
8	EVENT_NMBR	Char (30)	M	Running number after the prefix of the original transaction number from D365FO POS
9	REMARK	Char (30)	O	Reference of receipt number reprinted. For D365FO retail POS to send the Transaction type (1 char) + Running No. (5 chars) as RECEIPT_NMBR

5.3.3 CMD_CODE = 101 (Required)

This is the header record of each transaction.

Record Structure

Sno	Field Name	Type	M/O	Remark
1	RECEIPT_NMBR	Numeric (10)	M	The receipt number of this transaction. RECEIPT_NMBR cannot be duplicated within one transaction file. For D365FO retail POS to send the Transaction type (1 char) + Running No. (5 chars) as RECEIPT_NMBR
2	SHIFT_NMBR	Numeric (8)	M	Shift number from D365FO POS.
3	DATE	Date YYYYMMDD	M	Date when the sales transaction was performed.
4	TIME	Time HH:MM:SS	M	Time when the sales transaction was performed.
5	USER_ID	Char (20)	M	Tenant Staff ID who performed the operation.
6	MANUAL_RECEIPT	Numeric (10)	~~M~~ O	Standby receipt number (manual receipt number)
7	REFUND_RECEIPT	Numeric (10)	~~M~~ O	Original sales receipt number of a refund/return transaction. For a normal sales receipt, it must be NULL. If not NULL, it is interpreted as a refund/return transaction. For D365FO retail POS to send the Transaction type (1 char) + Running No. (5 chars) as RECEIPT_NMBR
8	REASON_CODE	Char (10)	~~M~~ O	Reason code of the refund/return transaction. NOT USE. It will be defined by item level in CMD 111.
9	SALESMAN_CODE	Char (20)	~~M~~ O	Salesman code. If there are multiple salesman within a transaction, the first salesman code will be used.
10	TABLE_NMBR	Char (10)	O	NOT USE (For F&B only)
11	CUST_COUNT	Numeric (2)	O	NOT USE (For F&B only)
12	TRAINING	Char (1) Y/N	M	Flag to determine whether training mode is on / off. For non production environment, this will be 'Y'.
13	TRAN_STATUS	Char (8)	M	"SALE" for retail POS sale.

Sno	Field Name	Type	M/O	Remark
				"SALES ORDER" for SO non-sales related transactions (e.g. Deposit Collection, SO Cancellation). "INVOICE" for SO sales related transactions (e.g. Deposit utilization, return).
14	TRANS_TYPE	Char (8)	M	Transaction type, possible values are : Null – Normal Sales; OL – Online Sales; DP – Duty paid; DF – Duty free; MX – Mixture of DP and DF;
15	EVENT_NMBR	Char (30)	M	Running number after the prefix of the original transaction number from D365FO POS
16	SO_NMBR	Char (20)	O	D365FO Sales order number related to the transaction.
17	SO_INVOICE_NMBR	Char (20)	O	D365FO Sales Invoice number related to the transaction.
18	TRANS_COMMENT	Char (60)	O	Transaction Comment

5.3.4 CMD_CODE = 102 (Optional)

This record stores the information of the customer for tenant additional customer info. If customer information is not entered, this record is not created.

Record Structure

Sno	Field Name	Type	M/O	Remark
~~1~~	~~PASSPORT_NMBR~~	~~Char(12)~~	~~O~~	~~Passport number of the customer. After PDPA changes, passport number will not be captured.~~
~~2~~	~~FLIGHT_NMBR~~	~~Char(10)~~	~~O~~	~~Flight number of the customer~~
~~3~~	~~NATION_CODE~~	~~Char(3)~~	~~O~~	~~Nationality code of the customer. If entered, it will validate against a pre-defined Nationality Master file. It is the same as the Nationality code in CMD_CODE 106. Detail refer to Annex – Nationality.~~
~~4~~	~~CONNECT_FLIGHT~~	~~Char(1)~~	~~O~~	~~Y/N~~ ~~Status to denote whether the customer is on a connecting flight~~
~~5~~	~~PAX_TYPE~~	~~Char(1)~~	~~O~~	~~Type of passenger~~ ~~A – Arrival~~ ~~D – Departure~~ ~~S – Transit~~ ~~T – Transfer~~ ~~O – Others~~
~~6~~	~~SEX~~	~~Char(1)~~	~~O~~	~~M – Male~~ ~~F – Female~~
~~7~~	~~AGE~~	~~Char(3)~~	~~O~~	~~Customer age.~~
1	CUST_INFO1	Char (20)	O	Tenant's loyalty member card number e.g. Member number:12345678 (Tenant Label:Value)
2	CUST_INFO2	Char (20)	O	Tenant's loyalty member name
3	CUST_INFO3	Char (20)	O	Reserved information field
4	CUST_INFO4	Char (20)	O	Reserved information field
5	CUST_INFO5	Char (20)	O	Reserved information field
6	CUST_INFO6	Char (20)	O	Reserved information field
7	CUST_INFO7	Char (20)	O	Reserved information field
8	CUST_INFO8	Char (20)	O	Reserved information field
9	CUST_INFO9	Char (20)	O	Reserved information field
10	CUST_INFO10	Char (20)	O	Reserved information field
11	CUST_INFO11	Char (20)	O	Reserved information field
12	CUST_INFO12	Char (20)	O	Reserved information field
13	CUST_INFO13	Char (20)	O	Reserved information field

Sno	Field Name	Type	M/O	Remark
14	CUST_INFO14	Char (20)	O	Reserved information field
15	CUST_INFO15	Char (20)	O	Reserved information field

5.3.5 CMD_CODE = 103 (Optional)

This record stores the information of the customer master. If customer master is not entered, this record is not created.

Record Structure

Sno	Field Name	Type	M/O	Remark
1	CUST_NAME	Chars (80)	M	Name of the customer.
2	PHONE_NMBR	~~Numeric(20)~~ Char (20)	O	Contact number of the customer.
3	ADDRESS_1	Chars (250)	O	Street of the customer address.
4	ADDRESS_2	Chars (250)	O	City, portal code & country of customer address.
5	EMAIL	Chars (100)	O	Email address of the customer.

5.3.6 CMD_CODE = 104 (Optional)

This record stores the order information for pickup/collection, e.g. deposit/pre-order. If such information is not entered, this record is not created.

Record Structure

Old version:

Sno	Field Name	Type	M/O	Remark
~~1~~	~~CUST_NAME~~	~~Chars(80)~~	~~M~~	~~Name of the customer.~~
~~2~~	~~IC_NMBR~~	~~Numeric(15)~~	~~O~~	~~NOT USE.~~ ~~Identification no. of the customer.~~
~~3~~	~~PHONE_NMBR~~	~~Chars(20)~~	~~O~~	~~Contact number of customer.~~
~~4~~	~~REMARK~~	~~Chars(100)~~	~~O~~	~~Remark from the order or customer.~~

New version:

Sno	Field Name	Type	M/O	Remark
1	PHONE_NMBR	~~Numeric(20)~~ Char (20)	O	Contact number of the customer.
2	ADDRESS_1	Chars (250)	O	Street of the customer address.
3	ADDRESS_2	Chars (250)	O	City, portal code & country of customer address.
4	EMAIL	Chars (100)	O	Email address of the customer.
5	COLLECTION_ DATE	Date YYYYMMDD	O	Collection date
6	COLLECTION_STORE	~~Char (20)~~ Char (100)	O	Collection store
7	Reserved 1	Char (20)	O	For future use
8	Reserved 2	Char (20)	O	For future use
9	Reserved 3	Char (20)	O	For future use

5.3.7CMD_CODE = 105 (Optional)

This record stores the *sales order* delivery mode information of the respective customer order at item level. If such information is not entered, this record is not created.

Record Structure

Old version:

Sno	Field Name	Type	M/O	Remark
~~1~~	~~ADDR_1~~	~~Chars(250)~~	~~O~~	~~Street of the customer address.~~
~~2~~	~~ADDR_2~~	~~Chars(100)~~	~~O~~	~~City, portal code & country of customer address.~~

New version:

Sno	Field Name	Type	M/O	Remark
1	ITEM_CODE	Char (16)	M	SKU code of the item
2	COLLECTION_ MODE	Char (20)	O	Pickup/Carry out
3	COLLECTION_DATE	Date YYYYMMDD	O	Collection date
4	COLLECTION_STORE	Char (10)	O	Collection store

5.3.8 CMD_CODE = 106 (Optional)

This record stores the information of the customer's passport. Each transaction may contain one of such record.

~~This information will not be captured from POS for JCAD and to be considered during CAG phase.~~

Record Structure

Sno	Field Name	Type	M/O	Remark
1	NAME	Char (50)	O	First name and last name of the customer Due to PDPA regulations, name is masked.
2	PASSPORT_NMBR	Char (12)	O	Number of the customer's passport. Due to PDPA regulations, passport number is masked.
3	NATION_CODE	Char (3)	O	Nationality of the customer (Based on ISO3166-1 alpha-3 code) Detail refer to Annex – Nationality.
4	DATE_OF_BIRTH	Date YYYYMMDD	O	Date of birth of the customer.
5	GENDER	Char (1)	O	Gender of the customer 'M' = Male 'F' = Female '-' = not specified

5.3.9 CMD_CODE = 107 (Optional)

This record stores the information of the customer's boarding pass. Each transaction may contain one of such record.

Record Structure

Sno	Field Name	Type	M/O	Remark
1	FLIGHT_NMBR	Char (10)	O	Flight number indicated on the boarding pass.
2	FROM_CITY	Char (12)	O	From City airport code
3	TO_CITY	Char(12)	O	To City airport code
4	DATE_OF_FLIGHT	Date	O	Date of flight indicated in the boarding pass
5	COMPARTMENT	Char (1)	O	Compartment code of the boarding pass (F=First class, J=Business class and Y=economy class)
6	SEAT_NMBR	Char (4)	O	Seat number indicated in the boarding pass.
7	NO_OF_LEGS	Char (1)	O	1,2,3 or 4. Minimum 1 segment. Maximum 4 segments to limit the size of the bar code
8	PASSENGER_NAME	Char (50)	O	Name of the passenger is masked.
9	ELECT_TICKET	Char (1)	O	Electronic ticket indicator. 'E': boarding pass issued against an electronic ticket.
10	PASSENGER_STATUS	Char (1)	O	Values are 0-9.
11	PASSENGER_DESC	Char (1)	O	Values are 0-9.
12	FREE_BAGGAGE_ALLOWANCE	Char (3)	O	Baggage allowance for tourist. E.g. 1K (kilo), 2L (pound) or 2PC (pieces).
13	CHECKIN_SOURCE	Char (1)	O	How the tourist is being checking. 'W' means through web.
14	BOARDINGPASS_ISSUANCE	Char (1)	O	
15	OPERATING_CARRIER_PNR_CODE	Char (7)	O	
16	FREQ_FLYER _AIRLINE_DESG	Char (3)	O	After PDPA, Frequent Flyer Airline Designation will not be captured This will be empty.
17	FREQ_FLYER_NUM	Char (16)	O	After PDPA, Frequent Flyer Number will not be captured This will be empty.
18	MARKETING_CARRIER_DESG	Char (3)	O	

5.3.10 CMD_CODE = 108 (Optional)

This record stores the information of the customer's email id for e-receipt (and/or marketing). If such information is not entered, this record is not created.

Record Structure

Old version:

Sno	Field Name	Type	M/O	Remark
~~1~~	~~PSC Number~~	~~Char(12)~~	~~O~~	~~PSC number of the customer~~

New version:

Sno	Field Name	Type	M/O	Remark
1	Email address	Char (100)	O	Customer email address
2	T&C	Char (10)	O	"Accept" or "Decline"
3	Reserved	Char (10)	O	Reserved field

5.3.11 CMD_CODE = 109 (Optional)

This record stores the non-traveler information if the purchaser is a non-traveler, e.g. airport staff, tenant staff, etc. If such information is not captured, this record is not created.

Record Structure

Old version:

Sno	Field Name	Type	M/O	Remark
~~1~~	~~ORGN~~	~~Char(50)~~	~~O~~	~~Organization of the staff.~~
~~2~~	~~IC_NMBR~~	~~Char(15)~~	~~O~~	~~Identification no. of the staff~~
~~3~~	~~APSTAFF_NAME~~	~~Char(50)~~	~~O~~	~~Name of the airport staff holder.~~
~~4~~	~~APSTAFF~~	~~Char(1)~~ ~~Y/N~~	~~O~~	~~Airport Staff Indicator.~~
~~5~~	~~RFIDINPUT~~	~~Char(1)~~ ~~Y/N~~	~~O~~	~~Y (from RFID), N (manual entry)~~

New version:

Sno	Field Name	Type	M/O	Remark
1	ID_TYPE	Char (20)	O	Identification type
2	ID_NMBR	Char (20)	O	Identification no.
3	AGE	Numeric (3)	O	Age (cashier to manually input)
4	FIELD_1	Char (20)	O	Reserved for future use
5	FIELD_2	Char (20)	O	Reserved for future use
6	FIELD_3	Char (20)	O	Reserved for future use

5.3.12 CMD_CODE = 110 (Optional)

This record stores the information of customers identified by CAG. If customer information is not entered, this record is not created.

Record Structure

Sno	Field Name	Type	M/O	Remark
1	CUST_INFO16	Char (20)	O	Strictly used by CAG
2	CUST_INFO17	Char (20)	O	Strictly used by CAG
3	CUST_INFO18	Char (20)	O	Strictly used by CAG
4	CUST_INFO19	Char (20)	O	Strictly used by CAG
5	CUST_INFO20	Char (20)	O	Strictly used by CAG
6	CUST_INFO21	Char (20)	O	Strictly used by CAG
7	CUST_INFO22	Char (20)	O	Strictly used by CAG
8	CUST_INFO23	Char (20)	O	Strictly used by CAG
9	CUST_INFO24	Char (20)	O	Strictly used by CAG
10	CUST_INFO25	Char (20)	O	Strictly used by CAG

5.3.13 CMD_CODE = 111 (Required)

This record stores information of the item sold including void items. If N different items are sold in a transaction, N records are created.

Record Structure

Sno	Field Name	Type	M/O	Remark
1	ITEM_CODE	Chars (16)	M	SKU code of the item
2	ITEM_QTY	Numeric (8,2) 99999999.99 or - 99999999.99	M	Quantity purchased. For refund, it is preceded by a negative (-) sign
3	ORG_PRICE	Numeric (20,2) 99999999.99	M	Original price of the product.
4	NEW_PRICE	Numeric (20,2) 99999999.99	M	Selling price of the item which can be changed by Price Override function. If NEW_PRICE = ORG_PRICE, it means there is no price override or department markdown. If ITEM_FLAG = blank and NEW_PRICE <> ORG_PRICE, it means department markdown is done to this item. If ITEM_FLAG = P and NEW_PRICE <> ORG_PRICE, it means price override is done to this item.
5	ITEM_FLAG	Char (1) V/P/Null	M	Status of this ITEM code V – void item, item value is not included in sales total P – price override Null value – normal item *Null value (nothing between 2 pipes for any middle field, or nothing between the last pipe and the record terminator for the last field)
6	TAX_CODE	Char (5)	O	This Tax code field will be assigned according to the transaction. For example: If sales transaction is applied **without tax** and item is flagged as “G”, the TAX_CODE = “N”; If sales transaction is applied **without tax** and item is flagged as “N”, the TAX_CODE = “N”; If sales transaction is applied **with tax** and item is flagged as “G”, the TAX_CODE = “G”; If sales transaction is applied **with tax** and item is flagged as “N”, the TAX_CODE = “N”;

Sno	Field Name	Type	M/O	Remark
				For detailed technical explanation, please refer to appendix 6.4
7	DISCOUNT_CODE	~~Char(15)~~ Char(20)	O	Discount code (only if used-defined discount code is used). If more than 1, show the first one only, detail will be shown in CMD151.
8	DISCOUNT_AMT	Numeric(20,2) 99999999.99 or -99999999.99	O	Total item discount amount given to this line item disregard of the value of ITEM_QTY. For refund, it is preceded by a negative (-) sign
9	RENTAL_DEPT	Char (20)	O	Department code for product based rental calculation.
10	SKU_CATG_TENANT	Char (20)	M	Category code of the item in tenant product hierarchy.
11	LABEL_KEYS	Char (3)	O	Combination of the label codes (Business Hour = Customer Type + Item Size). NOT USE (For F&B only)
12	ITEM_COMM	Char (1)	O	Flag to indicate whether item is subject to commission. NOT USE
13	ITEM_NSALES	Numeric (20,2) 99999999.99 or -99999999.99	M	Total net sales amount of this line item disregard of the value of ITEM_QTY: ITEM_QTY*NEW_PRICE – DISCOUNT_AMT – ITEM_STAX – Distributed Item Disc where Distributed Item Disc is the Overall Disc Amount (see CMD_CODE 121 DISCOUNT field) distributed to all discountable items, used for POS reports and POS Host backend. For refund, it is preceded with negative (-) sign.
14	DISCOUNT_BY	Numeric (20,2) 99999999.99 or -99999999.99	O	Amount of discount in % or $, for POS updating only. E.g.: If item disc is 20%, DISCOUNT_BY = 20 DISCOUNT_SIGN = % If item disc is $5.00 DISCOUNT_BY = 5 DISCOUNT_SIGN = $
15	DISCOUNT_SIGN	Char (1) % or $	M	Type of item discount, default to % % - percentage discount is given
16	ITEM_STAX	Numeric (20,2) 99999999.99 or -99999999.99	M	Item tax amount, for POS updating only. For refund, it is preceded
17	PLU_CODE	Char (80)	O	PLU code if user entered this code during sales
18	ORG_DUTY	Numeric (20,2) 99999999.99	O	Original custom + excise duty of the item. Applicable for Duty Free POS only.

Sno	Field Name	Type	M/O	Remark
19	NEW_DUTY	Numeric (20,2) 99999999.99	O	Custom + excise duty of the item which is changed by Price Override, equals to ORG_DUTY if no price override. Applicable for Duty Free POS only.
20	TOT_DUTY	Numeric (20,2) 99999999.99 or -99999999.99	O	Total custom + excise duty of the item, equals SKU_QTY * NEW_DUTY. Applicable for Duty Free POS only.
21	SKU_CATG_CAG	Char (20)	M	Category code of the item in JCAD/CAG product hierarchy.
22	ITEM_DESC	Char (60)		Description of the product.
23	DEPT_DESC	Char (60)	M	Category name of the item in CAG/JCAD product hierarchy.
24	CATG_DESC	Char (60)	M	Category name of the item in tenant product hierarchy.
25	ITEM_ATTRIB1	~~Char(20)~~ Char(60)	O	This is an item attribute for brand (To be defined by tenant).
26	ITEM_ATTRIB2	~~Char(20)~~ Char(60)	O	This is an item attribute for gender. Values: 'MALE','FEMALE','UNISEX'
27	ITEM_ATTRIB3	~~Char(20)~~ Char(60)	O	This is an item attribute for age group. Values: 'ADULT','CHILD<12', 'ALL'
28	ITEM_ATTRIB4	~~Char(20)~~ Char(60)	O	This is an item attribute for Changi collection. Values: 'CHANGIEXCL', 'LOVESG', 'TRAVELEXCL', 'TRAVELEXCL_LOVESG', 'CHANGIEXCL_LOVE SG', 'TRAVEL_CHANGI EXCL,'NA'
29	SALESPERSON_CODE	~~Char (8)~~ Char (20)	O	Salesperson ID of the line item.
30	SALES_TYPE	Char (40)	O	Sale type of item. e.g. ART_SALE: Normal Sale Item ART_SALESORDER: Sales order Item ART_INVOICE: SO Invoice Item ART_V_SET: Kit Item ART_RETURN: Return/refund item ART_LINE_VOID: Void Item ART_IMEDIATE_VOID: Void transaction
31	REASON_CODE	Char (10)	O	Reason code of the refund/return/void for line item.
32	DUMMYITEM_REF	Char (50)	O	This is record the SKU/barcode of the dummy item. Tenant to use the dummy item to register the sales when the SKU/barcode is not found in POS.
33	ITEM_COMMENT	Char (60)	O	This is record the product description of the dummy item.

Sno	Field Name	Type	M/O	Remark
34	ITEM_ATTRIB5	Char (60)	O	This is an item attribute for brand collection (To be defined by tenant). (E.g: series, limited edition). E.g: Lewis Hamilton X T H, Adidas NMD, GoPro Hero, iPhone
35	ITEM_ATTRIB6	Char (60)	O	This is an item attribute for Language. Values: 'ENGLISH', 'CHINESE', 'ALL OTHERS','NA' *This field is mandatory for selective tenants in the book concession.
36	ITEM_ATTRIB7	Char (60)	O	This is an item attribute for Genre. Values: 'K-COS', 'J-COS', 'ALL OTHERS','NA' *This field is mandatory for selective tenants in the perfume & cosmetics concession.
37	ITEM_ATTRIB8	Char (60)	O	This is an item attribute for Geography. Please strictly follow values stated below. Values: 'LOCAL', 'OVERSEAS', 'ALL' *This field is mandatory for selective tenants in the travel communication and iShopChangi concession.
38	ITEM_ATTRIB9	Char (60)	O	Reserved for future use.
39	ITEM_ATTRIB10	Char (60)	O	Reserved for future use.
40	ITEM_ATTRIB11	Char (60)	O	Reserved for future use.
41	ITEM_ATTRIB12	Char (60)	O	Reserved for future use.
42	ITEM_ATTRIB13	Char (60)	O	Reserved for future use.
43	ITEM_ATTRIB14	Char (60)	O	Reserved for future use.
44	ITEM_ATTRIB15	Char (60)	O	Reserved for future use.
45	DUMMYITEM_ DESC	Char (60)	O	This is the product description of the dummy item. Tenant to use the dummy item function to register sales when the SKU/barcode is not found in POS during transaction.
46	CONFIG	Char (10)	O	Product dimension Configuration.
47	SIZE	Char (10)	O	Product dimension Size.
48	COLOR	Char (10)	O	Product dimension Color.
49	STYLE	Char (10)	O	Product dimension Style.
50	UNIQUEID	Char (100)	O	Unique id captured during transaction. This applies for products which have the "Enable_UniqueID" flag = Yes.
51	TRANS_TYPE	Char (8)	O	DP – Duty paid DF – Duty free Null – Non duty
52	ORIG_CUSDTY	Numeric (20,2) 99999999.99	O	Original custom duty
53	ORIG_EXCDTY	Numeric (20,2)	O	Original excise duty

Sno	Field Name	Type	M/O	Remark
		99999999.99		
54	NEW_CUSDTY	Numeric (20,2) 99999999.99	O	New customer duty
55	NEW_EXCDTY	Numeric (20,2) 99999999.99	O	New excise duty

5.3.14 CMD_CODE = 115 (Optional)

This is the footer record of a deposit transaction. Note: Deposit is not considered as sales for rental calculation.

Record Structure

Sno	Field Name	Type	M/O	Remark
1	SHIFT_NMBR	Numeric (8)	M	Shift number from D365FO POS.
2	~~DEP_RECEIPT_NMBR~~ SO_NMBR	~~Numeric(10)~~ Char(20)	~~M~~ O	~~For D365FO retail POS to send the Transaction type (1 char) + Running No. (5 chars) as DEP_RECEIPT_NMBR~~ D365FO Sales order number related to the transaction.
3	~~DEP_REFUND_RECEIPT~~ TOTAL_SO_AMT	~~Numeric(10)~~ Numeric (20,2) 99999999.99 or - 99999999.99	~~M~~ O	~~Receipt number of the refund deposit transaction. For D365FO retail POS to send the Transaction type (1 char) + Running No. (5 chars) as DEP_REFUND_RECEIPT~~ Total SO Amount for the sales order.
4	DEP_AMOUNT	Numeric (20,2) 99999999.99 or - 99999999.99	M	Amount (cumulative) of the deposit transaction. For refund deposit entry, precede with negative (-) sign
5	FLAG	Char (1)	M	Flag of deposit entry C – Deposit Collection F – Deposit Forfeit R – Deposit Refund U – Deposit Utilization
6	ENTRY_DATE	Date YYYYMMDD	M	~~Date of the deposit transaction created.~~ SO created date.
7	~~EXPIRY_DATE~~ UTILIZATION_AMT	~~Date YYYYMMDD~~ Numeric (20,2) 99999999.99 or - 99999999.99	O	~~Expiry date of deposit entry.~~ Accumulated Deposit Utilization amount.
8	EMPL_ID	Char (20)	M	Tenant Staff ID who performed the deposit transaction.

Sno	Field Name	Type	M/O	Remark
9	TAX_CODE	Char (5)	M	Tax code used to compute tax amount for the deposit amount collected. For transaction with mixed item tax code, the first item tax code will be used.
10	TAX_AMT	Numeric (20,2) 99999999.99 or - 99999999.99	M	Tax amount computed. For refund, preceded with negative (-) sign. For transaction with mixed item tax code, this is the total tax amount regardless of the tax code.
11	TAX_TYPE	Char (5)	M	Type of tax: I – Deposit amount include tax E – Deposit amount exclude tax This is determined by the store on whether transaction is taxable or non-taxable.
12	TAX_EXEMPT	Char (5)	M	Tax exempt flag Y – Tax amount is exempted N – Tax amount is to be paid by customer
13	ADJUSTAMT	Numeric (20,2) 99999999.99 or - 99999999.99	O	Adjustment amount due to the 5 cents rounding.
14	DATE	Date YYYYMMDD	M	Date when the deposit transaction is performed.
15	TIME	Time HH:MM:SS	M	Time when the deposit transaction is performed.
16	SO_ACTION	Char (20)	O	D365FO Sales order Action triggered from POS: - Create - Edit - Pickup - Cancel

5.3.15 CMD_CODE = 121 (Required)

This is the footer record of a transaction that stores all Tax, Service Charge and Discount details. Service Charge are used for F&B only.

Record Structure

Sno	Field Name	Type	M/O	Remark
1	SALES	Numeric (20,2) 99999999.99 or - 99999999.99	M	Total sales amount before deducting discount. For refund, it is preceded with negative (-) sign. If prices are inclusive of GST, GST amount is included. If prices are exclusive of GST, GST amount is excluded.
2	DISCOUNT	Numeric (20,2) 99999999.99 or - 99999999.99	M	Discount given for this transaction. For refund, it is preceded with a negative (-) sign
3	CESS	Numeric (20,2) 99999999.99 or - 99999999.99	O	The cess for this transaction, for F&B POS only. CESS is **not** included in the SALES amount. For refund, it is preceded with negative (-) sign.
4	CHARGES	Numeric (20,2) 99999999.99 or - 99999999.99	O	Service charge for this transaction, for F&B only. CHARGES are **not** included in the SALES amount. For refund, it is preceded with negative (-) sign
5	TAX	Numeric (20,2) 99999999.99 or - 99999999.99	O	Tax amount collected or exempted for this transaction depending on flag EXEMPT_GST. For refund, it is preceded with negative (-) sign.
6	TAX_TYPE	Char (1)	M	Type of pricing method I – selling price inclusive of tax (applicable to landside/public) E – selling price exclusive of tax (applicable to airside)
7	EXEMPT_GST	Char (1) Y/N	M	Default to 'N'
8	DISCOUNT_CODE	Char (20)	M	Discount code used for receipt discount (if user-defined discount code is used)
9	OTHER_CHG	Numeric (20,2) 99999999.99 or - 99999999.99	O	Amount of other charges which is not subject to POS Host rental computation For F&B only.

Sno	Field Name	Type	M/O	Remark
10	DISCOUNT_PER	Numeric (20,2) 999999.99 or - 999999.99	O	Percentage of overall discount. If null and DISCOUNT is not zero, $ overall discount is given.
11	ADJUSTAMT	Numeric (20,2) 99999999.99 or - 99999999.99	O	Adjustment amount due to the 5 cents rounding. This amount should be added to the SALES amount for rental calculation purpose. For example, purchase amount is $12.72, rounding adjustment is -0.02. Customer will pay $12.70. If it is $12.73, rounding adjustment is +0.02. Customer will pay $12.75.
12	TOTAL_DUTY	Numeric (20,2) 99999999.99 or - 99999999.99	O	Total custom + excise duty amount. For refund, it is preceded with negative (-) sign. For DFS POS only.
13	QTY	Numeric (8,2) 99999999.99 or - 99999999.99	M	Total quantity of SKU items sold for this transaction.
14	RECEIPTAMT	Numeric (20,2) 99999999.99 or - 99999999.99	O	Amount for current receipt, regardless recall transaction.
15	SERVERSTAT	Char (1)	M	Indicates the transaction is done on Y - “ONLINE” or N - “OFFLINE” status. (Connection to RSSU Status)
16	ELAPSETIME	Numeric (5)		Period of a transaction been done (in seconds) Cart checkout time – First item scanned/added time
17	GROSS_SALES	Numeric (20,2) 99999999.99 or - 99999999.99	M	The total sales amount before deducting discount For refund, it is preceded with negative (-) sign. Applies when there are more than 1 tax type (tax_type, I or E) This field would not be generated if there is only 1 tax type.
18	BACKEND_UPD	Datetime YYYY-MM-DDTHH:MM:SS	M	This is a transaction record created based on the datetime at the retail backend

5.3.16 CMD_CODE = 131 (Required)

This record stores the currency code and amount tendered or changed or any rounding adjustment. Hence, a transaction can contain more than one of such record.

Record Structure

Sno	Field Name	Type	M/O	Remark
1	TYPE	Char (1)	M	Type of record T – Tender C – Change For positive adjustment in cmd121, it is C For negative adjustment in cmd121, it is T For refund, T & C will be reversed. For example, purchase amount is $12.72, rounding adjustment is -0.02 in cmd121 and T in cmd131. Customer will pay $12.70. If it is $12.73, rounding adjustment is +0.02 in cmd121 and C in cmd131. Customer will pay $12.75.
2	PAYMENT_NAME	Char (30)	M	For Cash: (E.g: Cash, Foreign currency) For Tenant vouchers: Type of the voucher (E.g: DBS, Safra). For Landlord vouchers: (E.g. CR eVoucher, CTP eVoucher, CDV) For cards: Card type (E.g: AMEX, VISA, MASTER, NETS) For Digital payment: E.g: AliPay, GrabPay) For Landlord gift card: (E.g: CAG Gift Card) For Tenant gift card: (E.g: Tenant gift card) For more details, refer to Annex – Payment media.
3	CURR_CODE	Char (3)	M	Currency code used
4	BUY_RATE	Numeric (12,7)	M	Exchange rate of the currency used
5	AMOUNT	Numeric (20,2) 99999999.99	M	Amount tendered / changed / adjusted Value is always without signage Amount for voucher should be equal to voucher_qty multiplied by voucher_value
6	REMARKS_1	Char (50)	O	Credit card number / cheque number / cash voucher number / NETS card number For credit card number, first
7	REMARKS_2	Char (50)		Authorization code for the credit card ~~or cash voucher quantity~~.
8	BASE_AMT	Numeric (20,2) 99999999.99	M	Amount of the local currency
9	USERPAYCODE	Char(10)	O	User-defined payment code.

Sno	Field Name	Type	M/O	Remark
				Payment code used by the tenant host system. For centralized voucher management, this will be retrieved from Payment voucher group "user pay code".
10	CC_NUMBER	Numeric (20)	O	Credit card number Passed from Credit Card/Nets Machine and manual entry
11	CEXPDATE	Char (4)	O	Credit card expiry date Passed from Credit Card/Nets Machine and manual entry
12	MERCHANTID	Char (15)	O	Merchant ID of credit card host / bank Passed from Credit Card/Nets Machine
13	APPR_CODE	Char (6)	O	Approval code return from the credit card terminal Passed from Credit Card/Nets Machine and manual entry
14	VOUCHER	Char (1)	O	Flag to indicate if this is a cash voucher payment Y – This is a cash voucher payment N – This is non-cash voucher
15	TRANSTYPE	Char (6)	O	Transaction type sends to the bank "SALES" – request to charge customer "VOID" – cancel the previous request Passed from Credit Card/Nets Machine
16	CCAINVNO	Numeric (6)	O	CCA invoice number That is the invoice number for credit/debit card transaction return from the Card Terminal Passed from Credit/debit Card
17	SUPVID	Char (20)	O	Staff ID who performs the CCA process Passed from Credit Card/Nets Machine
18	CREDPCDE	Char (84)	O	Credit card user pay code. Passed from Credit Card/Nets Machine
19	PAYMENT_NAMEGRP	Char (20)	O	Payment method group: 'Cash' 'Landlord voucher' 'Tenant voucher' 'Cards' 'Digital payment' 'CVM' 'Landlord gift card' 'Tenant gift card' 'Tenant e-voucher' 'Landlord e-voucher' 'Others'
20	DCC_CURR	Char (4)	O	Currency code of the DCC currency. Passed from Credit Card

Sno	Field Name	Type	M/O	Remark
21	DCC_RATE	Numeric (12,7)	O	Exchange rate of the DCC currency. Passed from Credit Card
22	VOUCHER_NO	Char (1000)	O	Voucher serial number, if voucher payment is used
23	PAY_ORIGIN	Char (20)	O	Payment origin: DEPOSIT – payment collected from Deposit Null value – payment (not from Deposit) *Null value (nothing between 2 pipes for any middle field, or nothing between the last pipe and the record terminator for the last field)
24	PAYOPT_GRP	Char (20)	O	Payment option group information (applicable only for Centralized Voucher Management (CVM), Landlord/Tenant defined payments), e.g. • For CVM – "Payment voucher group": ○ CAGCTPFLT0701 ○ Etc. • For Landlord defined payments – Group Id: ○ CAG • For Tenant defined payments – Group Id: ○ TNV ○ OTHERS • For e-Voucher (Landlord/Tenant), this will be = Payment name group.
25	PAYOPTGRP_DESC	Char (60)	O	Payment option group description (applicable only for CVM, Landlord/Tenant defined payments), e.g. • For CVM – Description: ○ CAG flight delay voucher • For Landlord defined payments – Group Id description: ○ CAG Vouchers • For Tenant defined payments – Group Id description: ○ Tenant Vouchers ○ Other payments • For e-Voucher (Landlord/Tenant), this will be = Payment name group .
26	PAYOPT _TYPE	~~Char (20)~~ Char (30)	O	Payment option type (applicable only for CVM, Landlord/Tenant defined payments), e.g. • For CVM – Type: ○ Landlord ○ Tenant • For Landlord defined payments – Type Id: ○ CDV ○ CTP ○ Etc.

Sno	Field Name	Type	M/O	Remark

				• For Tenant defined payments – Type Id:
				  ○ For Tenant Voucher:
				  ▪ DBS Voucher
				  ▪ Etc.
				  ○ For Other payment:
				  ▪ Wire Transfer
				  ▪ Giro
				  ▪ Etc.
				• For e-Voucher (Landlord/Tenant), this will be = "*e-Voucher Type*" from the API respond (if provided).

27	PAYOPT_NAME	~~Char (60)~~
		Char (150)
			O	Name of the payment option type (applicable only for CVM, landlord/tenant defined payments), e.g.
				• For CVM – Voucher pay name (Receipt):
				  ○ Landlord
				  ▪ CTP VOU
				  ▪ Etc.
				  ○ Tenant
				  ▪ DFS VOU
				  ▪ Etc.
				• For Landlord defined payments – Type Id Description:
				  ○ Changi Dollar Voucher
				  ○ CTP Voucher
				  ○ Etc.
				• For Tenant defined payments – Type Id Description:
				  ○ For Tenant Voucher:
				  ▪ DBS Voucher
				  ▪ Etc.
				  ○ For Other payment:
				  ▪ Wire Transfer
				  ▪ Giro
				  ▪ Etc.
				• For e-Voucher (Landlord/Tenant), this will be = "*e-Voucher Type Description/Name*" from the API respond (if provided).

28	PAYOPT _QTY	Char (10)	O	Payment option quantity: • For CVM, it will always be populated as "1". • For Landlord/Tenant defined payments, it will be populated as per the payment quantity defined. • For e-Voucher (Landlord/Tenant), this will be populated will the voucher quantity accordingly.

Sno	Field Name	Type	M/O	Remark
29	PAYOPT_VALUE	Numeric (20,2)	O	Payment option value: • For CVM, it will be as per the voucher serial number value. • For Landlord/Tenant defined payments, it will be populated as per the payment value defined. • For e-Voucher (Landlord/Tenant), this will be populated with the voucher value accordingly.

5.3.17 CMD_CODE = 141 (Optional)

This is the footer record of a transaction that stores the tax details. If two tax codes are used for the items sold, two records are created.

Record Structure

Sno	Field Name	Type	M/O	Remark
1	TAX_CODE	Char (1)	M	Tax code
2	AMOUNT	Numeric (20,2) 99999999.99	M	Tax amount paid for this tax code, may be zero.

5.3.18 CMD_CODE = 145 (Optional)

This record stores the amount from an accumulated transaction. Hence, a recall transaction can contain more than one of such record.

Record Structure

Sno	Field Name	Type	M/O	Remark
1	POSNO	Char (6)	O	POS ID
2	RECEIPTNO	Numeric (10)	O	Accumulated transaction receipt number (from other receipt number). For D365FO retail POS to send the Transaction type (1 char) + Running No. (5 chars) as RECEIPT_NMBR
3	PAYAMT	Numeric (20,2) 999999999 .99		Paid amount from an accumulated transaction (from other receipt number).
4	EVENT_NMBR	Numeric (30)	O	Running number after the prefix of the original transaction number from D365FO POS

5.3.19 CMD_CODE = 151 (Optional)

This is the footer record of a transaction that stores additional information needed for user-defined item, redeemed coupon and periodic discount. E.g. Discount code 1 is triggered by centralized coupon management. The coupon number entered is recorded in this command code for information.

Record Structure

Sno	Field Name	Type	M/O	Remark
1	DISCOUNT_CODE	Char (20)	O	User-defined discount coupon code used.
2	DISCOUNT_INFO	Char (60)	O	Description of the discount coupon used.
3	SYS_DISC_CODE	Char (20)	O	System discount code
4	CAMPAIGN_CPN_SERIALNO	Char (20)	O	Campaign coupon serial number used to trigger the discount.
5	CAMPAIGN_CPN_GRP	Char (20)	O	Coupon identification, e.g. XMAS2019$2OFF
6	CAMPAIGN_CPN_GRP_DESC	Char (60)	O	Description of the coupon given.
7	CAMPAIGN_CODE	Char (20)	O	Campaign code used to trigger the discount. E.g. XMAS2019
8	CAMPAIGN_DESC	Char (60)	O	Name of the campaign.
9	CAMPAIGN_GRP	Char (20)	O	Campaign group used to trigger the discount, E.g. XMAS2019
10	CAMPAIGN_GRP_NAME	Char (60)	O	Name of the campaign group.
11	CAMPAIGN_TYPE	Char (20)	O	Type of campaign, e.g. - Landlord - Tenant
12	LEGAL_ENTITY	Char (20)	O	Legal entity defined in the campaign code master (this is applicable for *Campaign_type* = Tenant)
13	ITEM_CODE	Char (16)	O	SKU code of the item
14	DISCOUNT_AMT	Numeric (20,2)	O	Discount amount.
15	DISCOUNT_TYPE	Char (60)	O	Type of discount E.g. Mix and match discount, Threshold discount,
16	CONCURRENCY_MODE	Char (60)	O	Discount concurrency mode, e.g. Exclusive, Best price, Compounded
17	PRICING_PRIORITY	Char (20)	O	Pricing priority from discount code master

5.3.20 CMD_CODE = 152 (Optional)

This is the footer record of a transaction that stores the campaign or coupon giveaway/receipt.

Record Structure

Sno	Field Name	Type	M/O	Remark
1	CAMPAIGN_ CPN_GRP	Char (20)	O	Coupon identification, e.g. XMAS2019$2OFF
2	CAMPAIGN_CPN_ GRP_DESC	Char (60)	O	Description of the coupon given.
3	CAMPAIGN_CODE	Char (20)	O	Campaign code for the giveaway. E.g: XMAS2019
4	CAMPAIGN_DESC	Char (60)	O	Description of the campaign.
5	CAMPAIGN_GRP	Char (20)	O	Campaign group that trigger the giveaway.
6	CAMPAIGN_GRP_ NAME	Char (60)	O	Name of the campaign group.
7	CAMPAIGN_TYPE	Char (20)	O	Type of campaign, e.g. - Landlord - Tenant
8	LEGAL_ENTITY	Char (20)	O	Legal entity defined in the campaign code master (this is applicable for *Campaign_type* = Tenant)
9	PAYOPT_GRP	Char (20)	O	Payment option group information (applicable only for Centralized Voucher Management (CVM), ~~Landlord/Tenant defined payments~~), e.g. • For CVM: ○ CAGCTPFLT0701 ○ Etc. • ~~For Landlord defined payments:~~ ○ ~~LLDVOU~~ • ~~For Tenant defined payments:~~ ○ ~~TNTVOU~~ ○ ~~OTHERS~~
10	QUALIFIED_TRX_A MT	Numeric (20,2)	O	Campaign/Coupon qualified transaction amount. E.g. Total receipt amount = $230, Tobacco amount (excl.) = $30, so the value shown here = $200.

5.3.21 CMD_CODE = 161 (Optional)

This record stores the loyalty customer information if the purchaser is a loyalty customer. If such information is not captured, this record is not created.
(This is only applicable for Changi Rewards)

Record Structure

Sno	Field Name	Type	M/O	Remark
1	CARD_NMBR	Char (50)	M	Card number captured at the POS. It will show first 19 character
2	BONUS_POINT	Numeric (20,2) 9999999999.99 or -9999999999.99	O	Bonus points recorded at the POS. For a refund transaction, this field should be negative.
3	REDEEMED_POINT	Numeric (20,2) 9999999999.99	O	Redeemed bonus points recorded at the POS.
4	CR_STAT	Char (1)	M	Indicates the transaction to Changi Rewards system is done on Y - “ONLINE” or N - “OFFLINE” status.
5	QUALIFIED_AMT	Numeric (20,2)	M	It can be zero. E.g. purchase $100, pay by voucher $20, so the qualified amount is $80 (need to exclude restricted product, if any).

5.3.22 CMD_CODE = 162 (Optional)

This record stores the integrated Changi Pay payment details integrated from the "Create Bill – Response" payment service provider's API.

A transaction can contain more than one such record.

If such information is not captured, this record is not created.

(This is only applicable for payment by ChangiPay via the integrated digital payment process on the POS)

Record Structure

Sno	Field Name	Type	M/O	Remark
1	LineNumber	Char (2)	O	Sequence number of Changi Pay payment line
2	Type	Char (30)	O	Payment type, provided by the payment service provider's API (e.g. "Credit Card", "CHANGI eVOUCHERS", "Merchant eV")
3	TypeCode	Char (30)	O	Payment type code, provided by the payment service provider's API (e.g. "eVoucher5-2020", "null")
4	TypeName	Char (100)	O	Payment type name, provided by the payment service provider's API (e.g. "CR2020 Changi Rewards $5 e-Voucher", "null")
5	Mode	Char (20)	O	Payment mode, provided by the payment service provider's API (e.g. "104")
6	Value	Numeric (18 , 2)	O	Payment value in currency code, provided by the payment service provider's API
7	CurrencyCode	Char (3)	O	Currency code, provided by the payment service provider's API (e.g. "SGD")
8	VoucherNo	Char (60)	O	Voucher number, provided by the payment service provider's API (e.g. "eVoucher5-202014428", "be70163e-bde0-41f1-83d0-12513885e503 ", "null")
9	DiscountID	Char (20)	O	Discount ID , provided by the payment service provider's API (e.g. "371")
10	Quantity	Numeric (1)	O	Always 1

6 ANNEX

All the list in Annex are subjected to changes.

6.1 Nationality

Nationality code	Country
10	AMERICA
11	RUSSIA
21	CHINA
22	INDIA
23	INDONESIA
24	JAPAN
25	KOREA
26	MALAYSIA
27	TAIWAN
28	THAILAND
29	PHILIPPINES
31	SINGAPORE
32	HONG KONG
33	MIDDLE EAST
34	MYANMAR
35	CAMBODIA
36	BANGLADESH
37	LAOS
38	SRI LANKA
40	EUROPE
50	AUSTRALIA
60	AFRICA
70	VIETNAM
71	BRUNEI
72	NEW ZEALAND
73	SWITZERLAND
88	OTHERS

6.2 Payment Media

Payment Name Group	Payment Media
Cash	Cash, Foreign currency.
Cards	NETs, VISA, MASTER, AMEX, DINERS, JCB, UNIONPAY
Landlord voucher	Changi voucher, etc.
Tenant voucher	Bank voucher (DBS, CitiBank & etc.), Safra, tenant brand voucher & miscellaneous.
Digital payment	AliPay, WeChatPay, etc.
CVM	Centralized payment voucher management
Landlord gift card	CAG (Changi rewards) gift card
Tenant gift card	Tenant gift card
Tenant e-voucher	Tenant e-voucher
Landlord e-voucher	Landlord (Changi rewards) e-voucher
Others	Other payments, e.g. Wire transfer, Giro, etc.

6.3 CAG/Jewel Product Hierarchy (sample glossary)

Please refer to the Interface TXT Format Excel file, “SKU Hierarchy” tab.

6.4 CMD 111 TAX_CODE matrix table

TAX_CODE value is derived by computing and populating in the POS based on 2 variables - Store Tax Group Setup and Item Tax Group Setup, as below.

Stores	Taxable	Store Tax Group	Item Tax Group / SKU Tax Code	TAX_CODE
LANDSIDE	Yes	G	G N	G N
LANDSIDE (NON-GST)	No	N	G N	N N
AIRSIDE	No	N	G N	N N
AIRSIDE (GST MODE)	Yes	G	G N	G N